ME1103
Principles of Mechanics and Materials

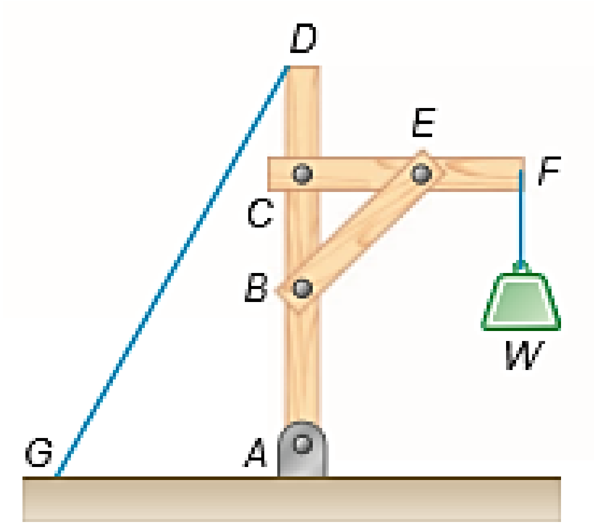
STATICS ANALYSIS OF TRUSS

Learning Outcomes

- Understand the *characteristics of truss*.
- Apply the *concept of static equilibrium* in solving *reaction forces at supports*.
- Understand the *methods of joints and sections* and apply them in truss analyses to determine the forces in truss members.

Engineering Structures

- Engineering structures are often constructed with *several connected parts*, such as in the following crane structure.



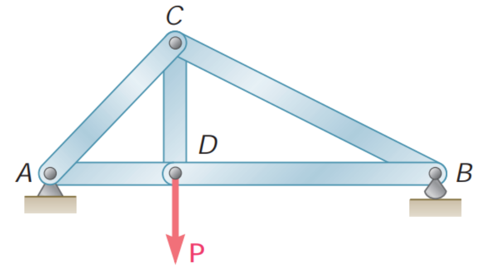
- Can this crane withstands the applied load *W*?
  - What material and their dimensions should be used for the rods?
  - What should be the size for the pins?
  - How thick should the cable be?
  - …

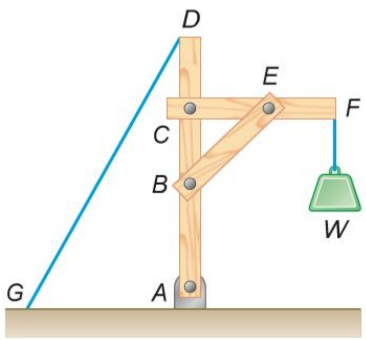
- Failure can often only be assessed from stress analysis (to be covered in Part II by Prof Vincent), *NOT* from force and moment via static analysis.
- But solving for the forces and moments in the assembly is the *pre-requisite*.

Engineering Structures...

- In this course, we are only going to focus on *pin-jointed structures*, in which the pins can *only constrain translation but not rotation* of the connected members. This also implies that the pins can *only experience forces* imposed by the members.

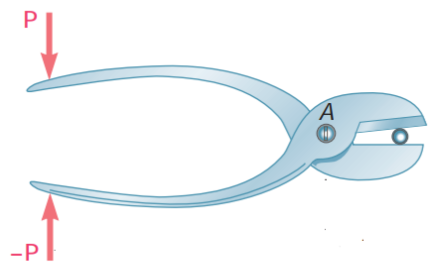
Trusses – usually designed to support static loads in fully constrained structures. They consist *exclusively* of *straight members* connected at joints at the ends.





Frames – They are usually rigid structure to support loads, with at least one of the members is a *multi-force member*.

Machines – designed to transmit and modify forces. They also have at least one *multi-force member* and always contain *moving parts*.



Truss in Engineering

- Truss is a major engineering structure in the design of bridges and buildings.

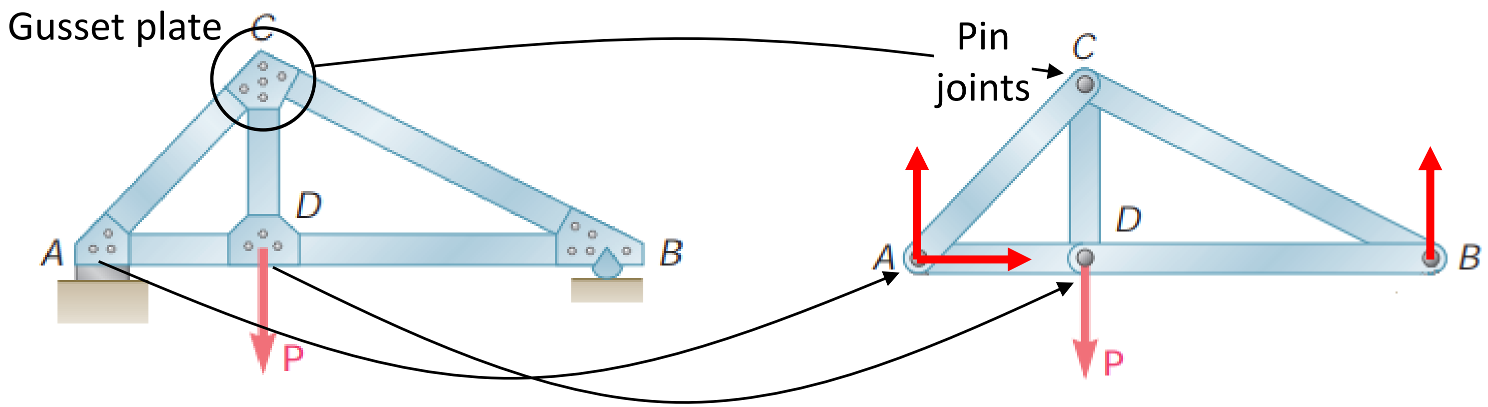






Definition of a Truss

- Truss consists of *straight members* connected *at the ends by pin joints* only. Hence, they are *two force members*, which means that the members can only withstand *axial force*.

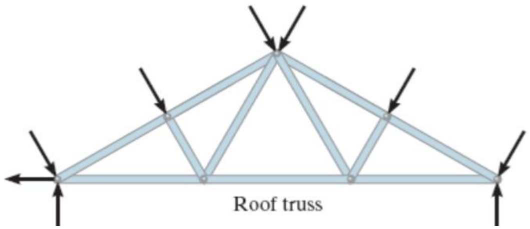


- All the *external loadings*, including the reaction forces at the supports, are assumed to be at *applied at the pin joints* only.

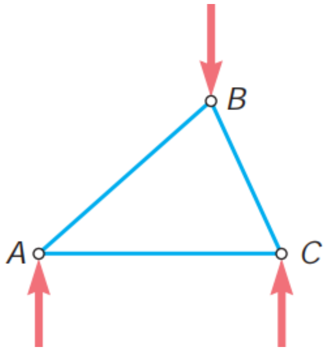
- Note that if distributed loads are applied on the members, e.g. weight of members, they will be converted into equivalent point forces to be applied onto the pin joints.

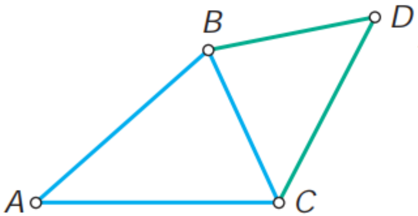
Definition of a Truss...

- A truss is often designed to carry loads which act is its plane and thus may be treated as a *two-dimensional structure*. They are referred to as *plane truss*.



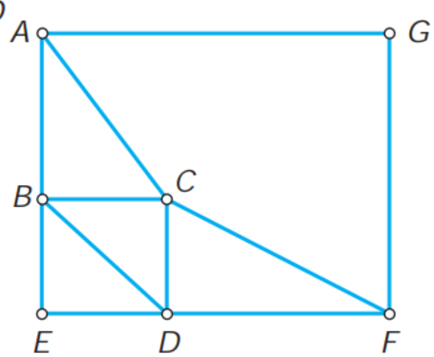
- The simplest design of a truss is formed by the basic *triangular truss*, with *two members to one connection*.





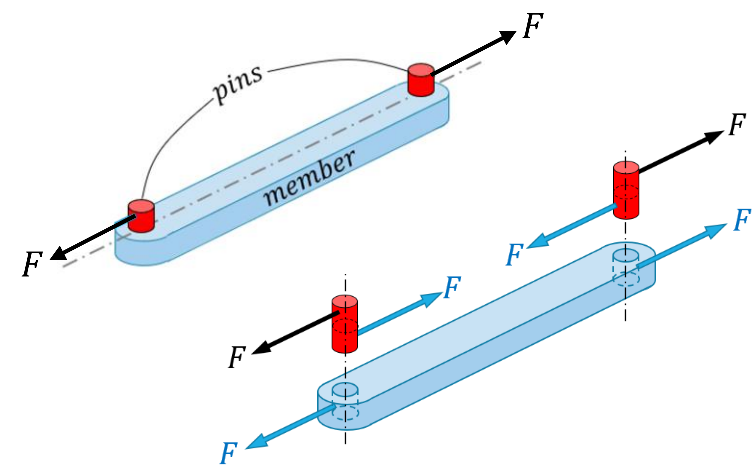
Simple truss

$m = 2n - 3$

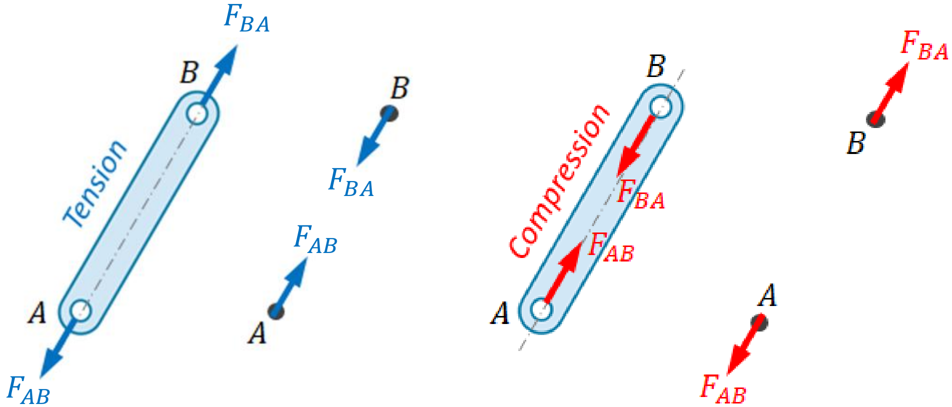


Definition of a Truss…

- Recall that pin joint can only restrain translation motion on the connecting body. This implies that it can only support forces.



- Since the forces can only act at the two pin joints, the truss member is a ***two-force member***, and hence the ***line of action*** must be along the member axis.



- The forces acting on the member by the pins, and the forces acting on the pins by the member internal force are ***action-reaction pairs***.

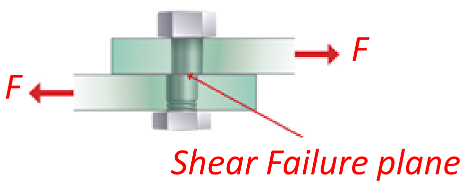
- The truss members can only be either in ***Tension***, ***Compression*** or ***Load free*** state.

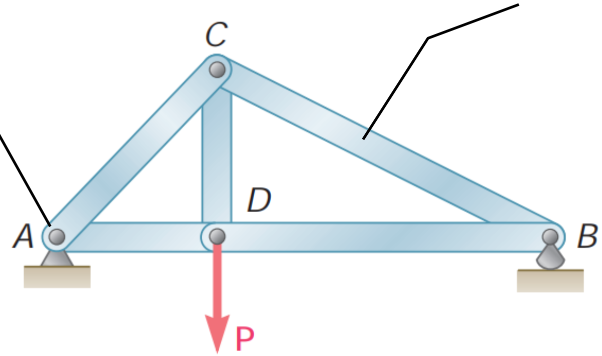
Analysis of Truss – Why and How?

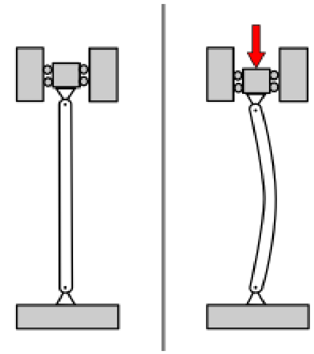
- For a truss, it is important to determine the forces acting on the pins and members, which is in turn used to check whether the truss can withstand the applied loads.

Can the pin withstand the shear load?

Is the member in tensile of *compression* state?





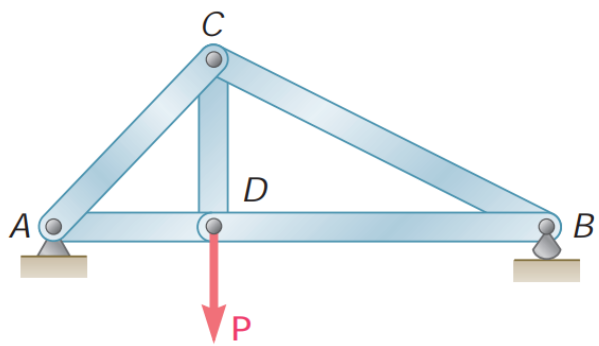


Buckling of rod under compressive axial load

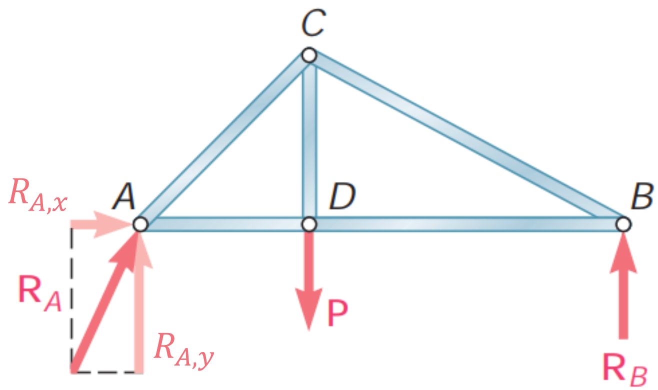
How to determine the forces in truss members and joints?

Method of joints **Method of sections.**

Reaction Forces at Supports



FBD of the entire truss



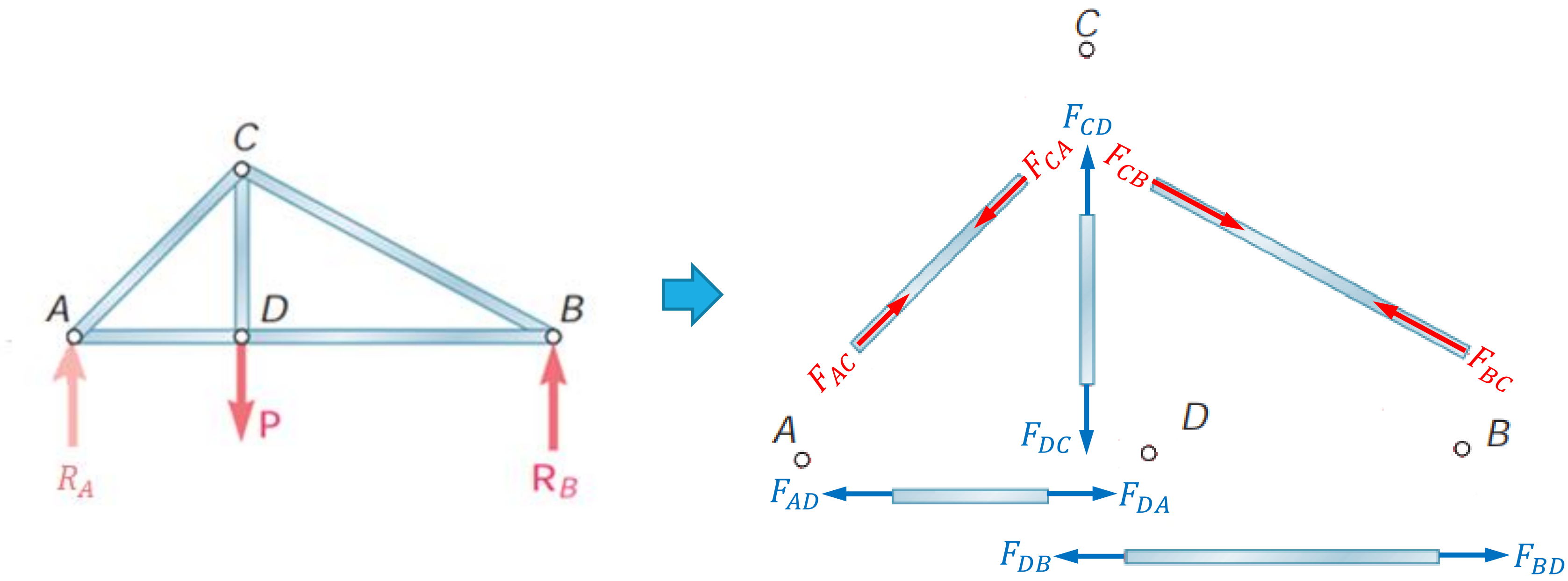
- Analyze the static equilibrium of the *entire truss* to get the reaction forces at the supports, i.e.

$$\sum F_x = 0\,, \qquad R_{A,x} = 0$$

$$\sum M_A = 0\,, \qquad R_B = \frac{AD}{AB}P$$

$$\sum M_B = 0\,, \qquad R_{A,y} = \frac{BD}{AB}P$$

Method of Joints

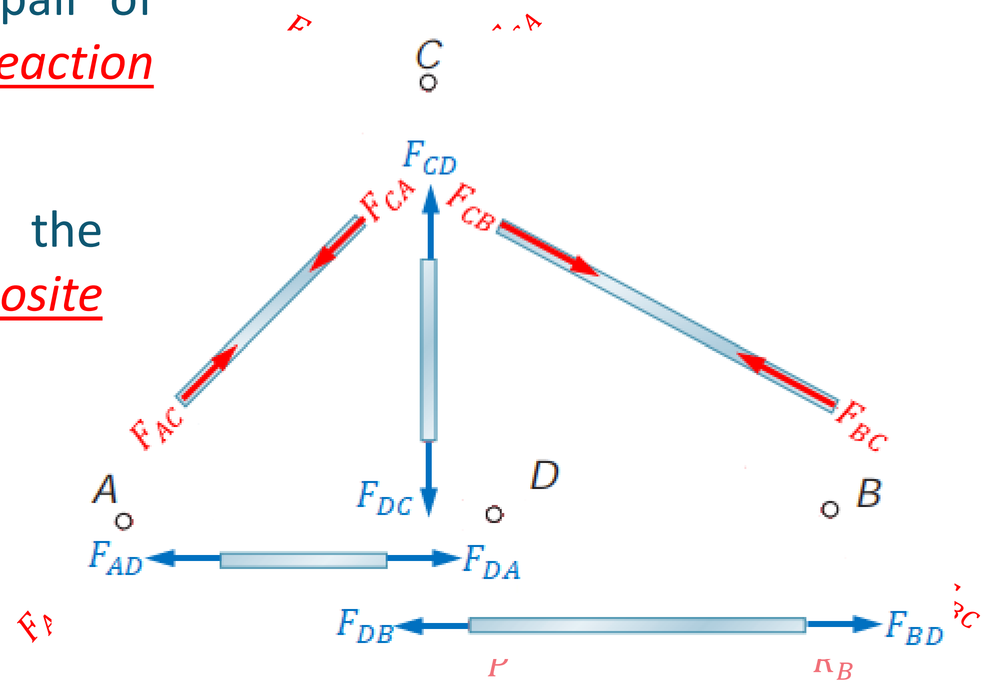


- Dis-membered the truss into the various components and draw a free body diagram for each pin and member.

- Recall that the truss members are ***two-force body***. Hence, the forces at the two ends of each member are equal and opposite, and the ***line of action is along the axis of the member***.

Method of Joints…

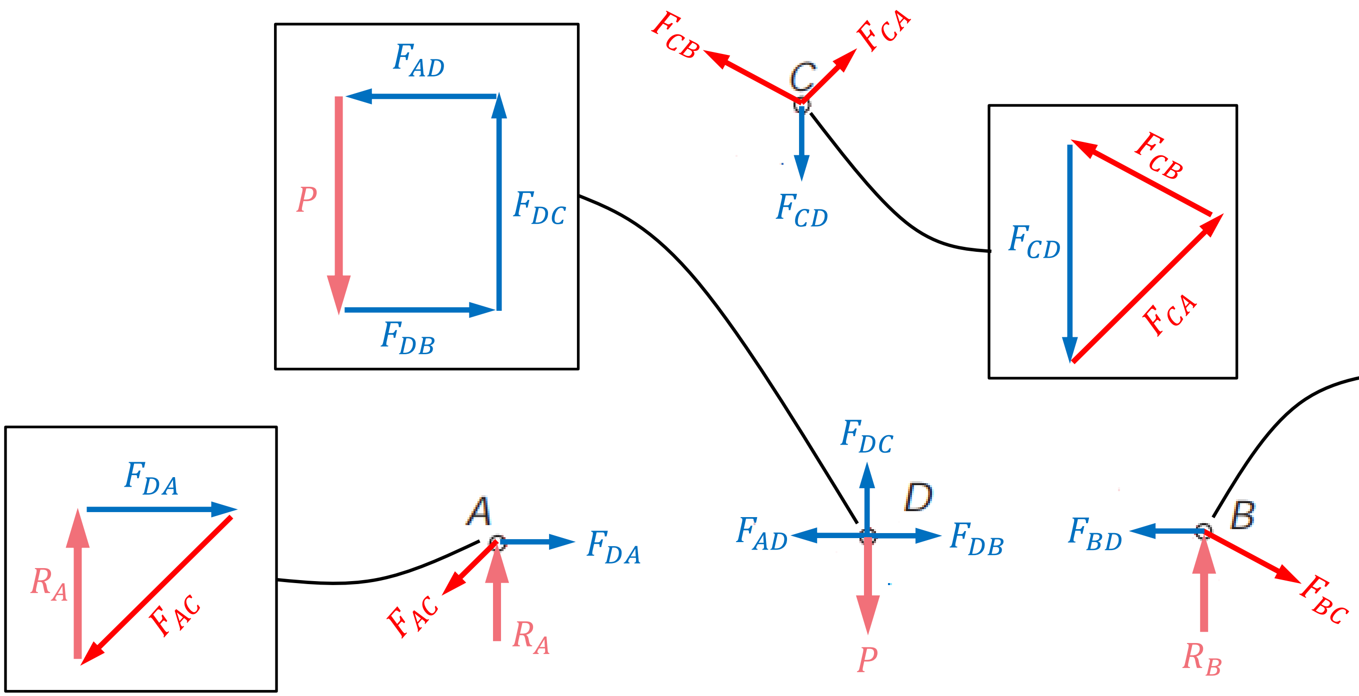
- The internal forces acting on each pair of pins and members are ***action and reaction pair***.
- This means that the directions of the forces acting on the pins must be ***opposite*** to that acting on the members.
- Notice that all the forces are ***concurrent*** at the joints, and hence force static equilibrium conditions must be satisfied.



- Hence, the ***Method of Joints*** is about conducting ***force static analyses*** on the joints ***sequentially***.

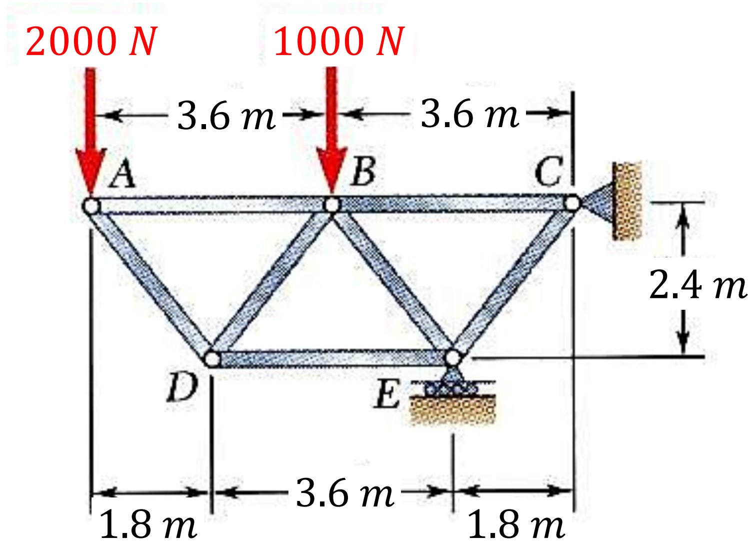
Method of Joints…

- Start with the joints with <u>*only two unknowns*</u>, i.e. either joint *A* or *B*. This is because there are only two force equations at each joint.



- We need only analyze 3 of the joints (6 equations) to solve for the unknowns forces in the 5 members. The extra joint analysis can be used for checking.

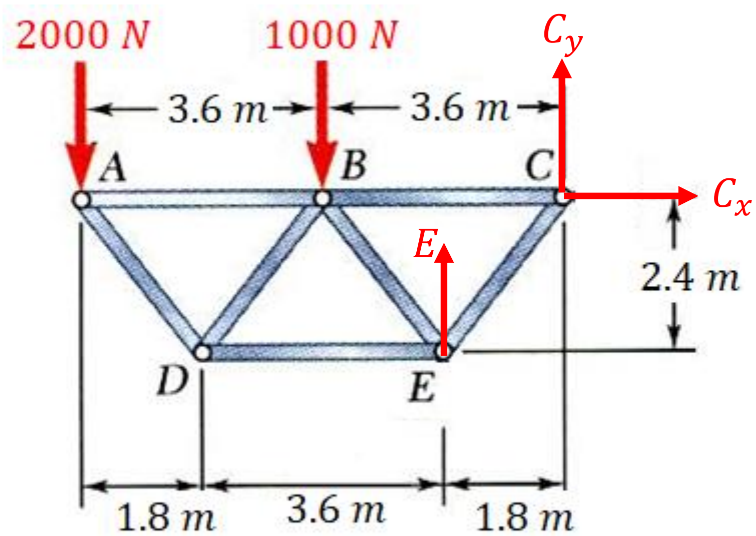
Example 3.1



Using the method of joints, determine the force in each member of the truss shown.

Example 3.1…

Free-body diagram



First, taking moment about C, gives

$$(2000)(7.2) + (1000)(3.6) - 1.8(E) = 0$$

$$\Rightarrow E = 10000\ N$$

Next, base on the force equilibrium, we have

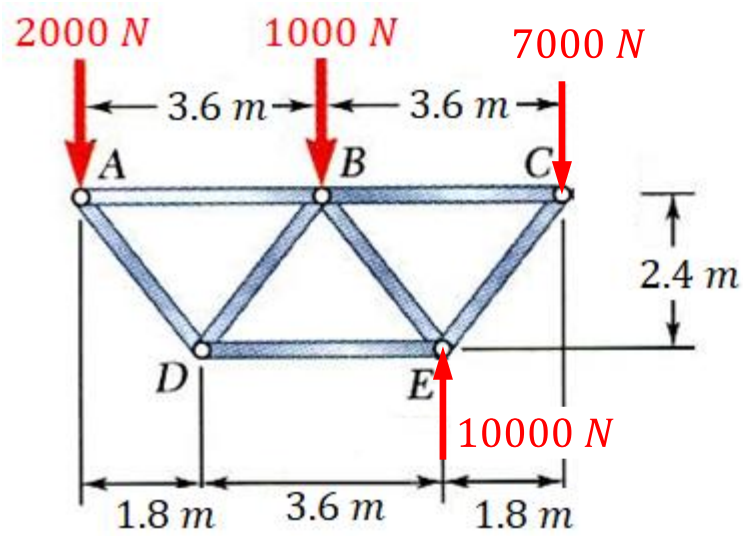
$$C_x = 0$$

And

$$-2000 - 1000 + E + C_y = 0$$

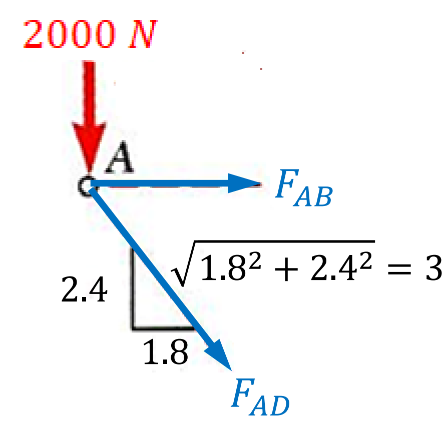
$$\Rightarrow\ C_y = -7000\ N$$

Example 3.1…



Next, we look for joints with only two unknown forces, and then apply the force equilibrium conditions to determine the forces.

Consider joint *A*



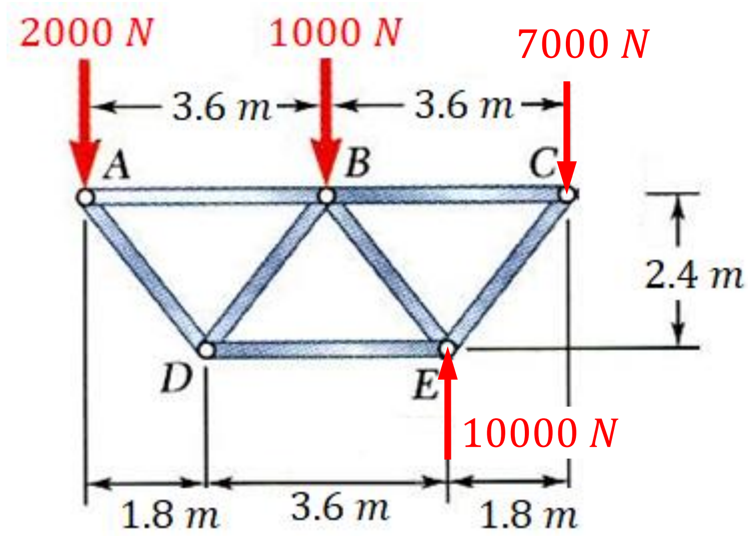
Force equation in the vertical direction, gives

$$2000 + F_{AD}\left(\frac{2.4}{3}\right) = 0 \Rightarrow F_{AD} = \mathbf{-2500\ N}\ (compression)$$

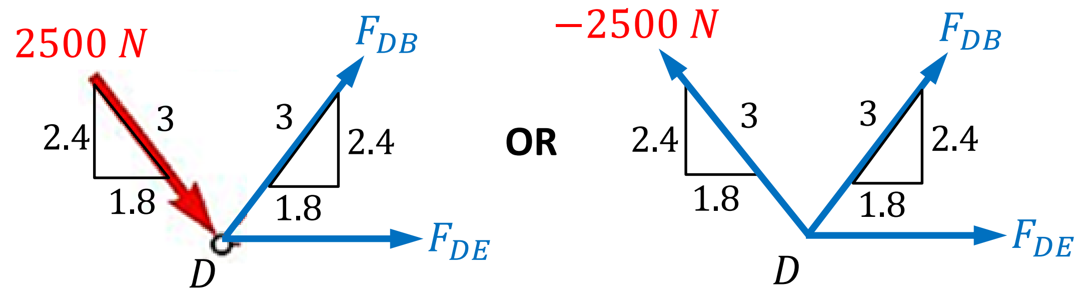
And force equation in the horizontal direction, gives

$$F_{AB} + F_{AD}\left(\frac{1.8}{3}\right) = 0 \Rightarrow F_{AB} = \mathbf{1500\ N}\ (tension)$$

Example 3.1...



Next, consider joint D



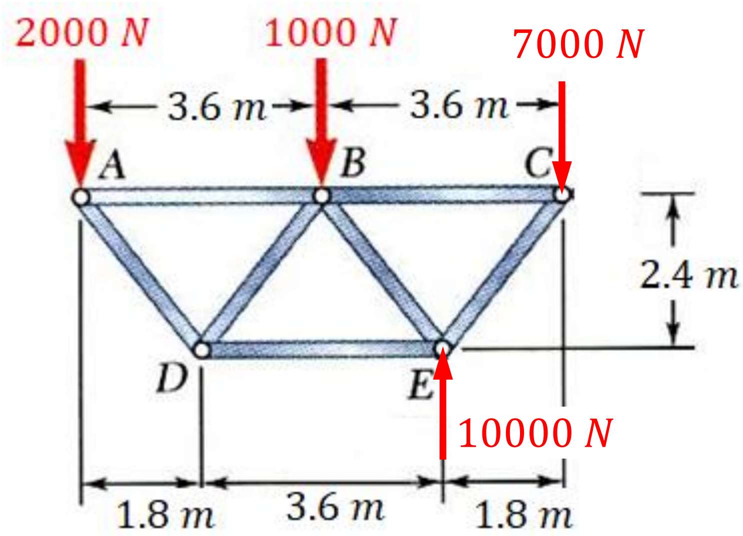
Based on the vertical force equation, we have

$$\left(\frac{2.4}{3}\right)F_{DB} - \left(\frac{2.4}{3}\right)(2500) = 0 \Rightarrow F_{DB} = F_{DA} = \mathbf{2500\ N}$$

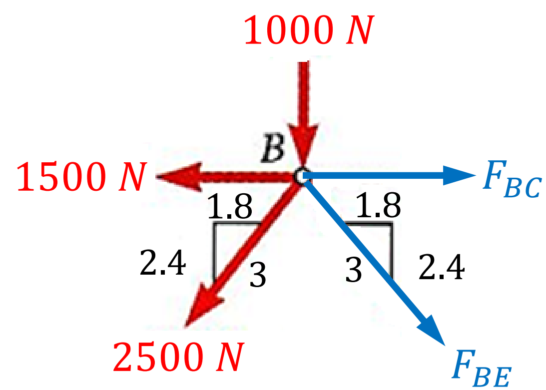
And the horizontal force equation gives

$$F_{DE} + 2\left(\frac{1.8}{3}F_{DA}\right) = 0 \Rightarrow F_{DE} = \mathbf{-3000\ N}$$

Example 3.1...



Consider joint B, we have

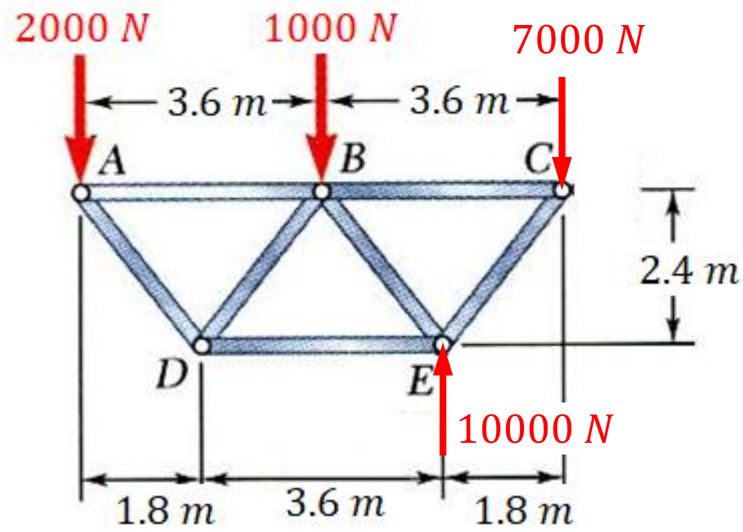


Again from force equilibrium equations,

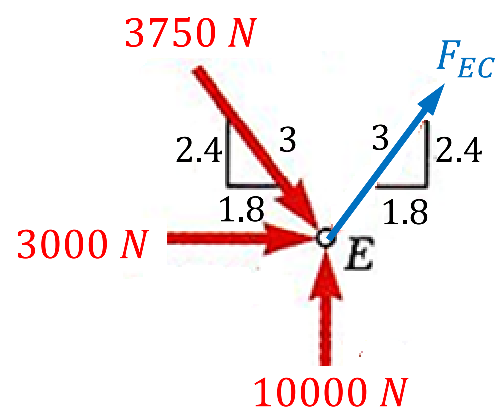
$$-1000 - \frac{2.4}{3}(2500) - \frac{2.4}{3}(F_{BE}) = 0 \Rightarrow F_{BE} = \mathbf{-3750\ N}$$

$$F_{BC} - 1500 - \frac{1.8}{3}(2500) - \frac{1.8}{3}(3750) = 0 \Rightarrow F_{BC} = \mathbf{5250\ N}$$

Example 3.1…



Finally at joint E, we have

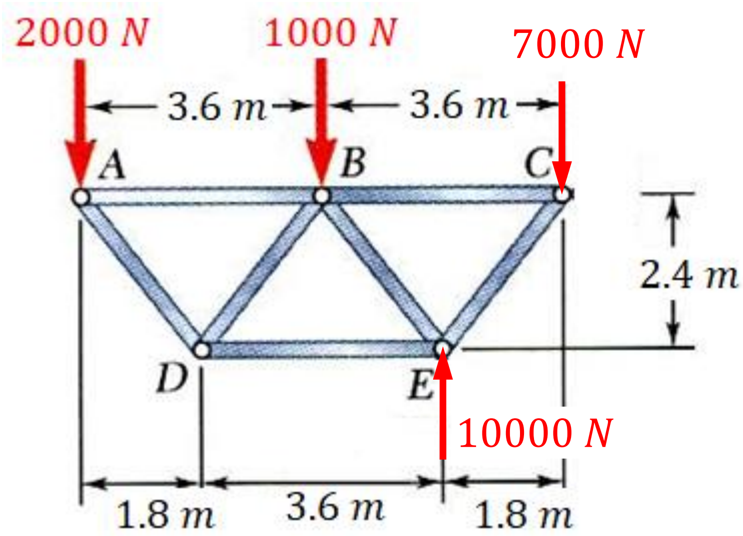


Using either the horizontal or vertical force equation, we get

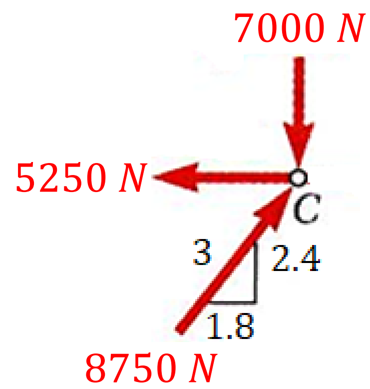
$$\left(\frac{1.8}{3}\right)F_{EC} + 3000 + \left(\frac{1.8}{3}\right)(3750) = 0 \Rightarrow F_{EC} = \mathbf{-8750\ N}$$

$$10000 - \left(\frac{2.4}{3}\right)(3750) + \left(\frac{2.4}{3}\right)F_{EC} = 0 \Rightarrow F_{EC} = \mathbf{-8750\ N}$$

Example 3.1...



Although all the forces in the members were found, we can use joint C as a check.



So according to the force equilibrium condition, we should get

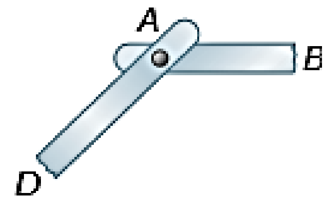
$$\sum F_x = -5250 + \left(\frac{1.8}{3}\right) 8750 = 0 \ (checked)$$

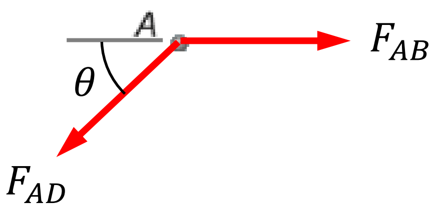
$$\sum F_y = -7000 + \left(\frac{2.4}{3}\right) 8750 = 0 \ (checked)$$

Method of Joints – Zero-force members

- Method of joints is straightforward but laborious, if the truss has many joints.
- Now, there exist three special configurations that can give rise to ***zero-force members***, which can be ***removed*** from the truss to simplify the analysis.

Case 1: Consider a joint with only ***two members*** with ***no external loading***, and they are ***not aligned*** to each other.





Force equilibrium along the horizontal direction, gives

$$F_{AD}cos\theta = F_{AB}$$

But in the vertical direction,

$$F_{AD}sin\theta = 0 \Rightarrow F_{AD} = 0$$

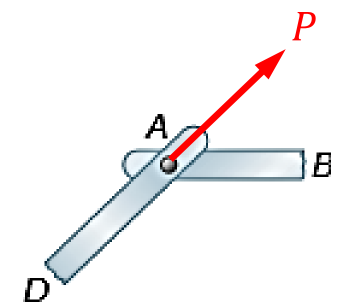
This also implies that $F_{AB} = 0$.

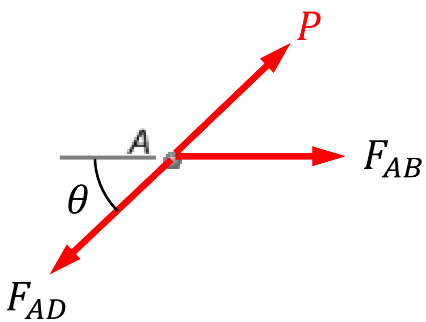
- Hence, ***both the members*** are zero-force members.

Method of Joints – Zero-force members…

Case 2: Consider a joint with only *two members* and they are *not aligned* to each other.

And suppose an external load *P* is applied on the joint and its line of action is *aligned with* one of the member.





Force equilibrium along to the line of action of *P* gives

$$P - F_{AD} + F_{AB}cos\theta = 0$$

And force equilibrium in the perpendicular direction gives

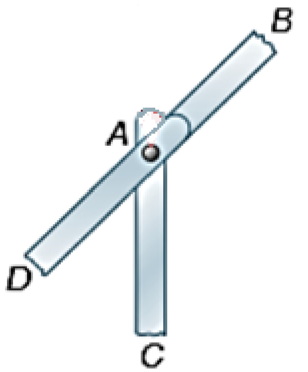
$$F_{AB}sin\theta = 0 \Rightarrow F_{AB} = 0$$

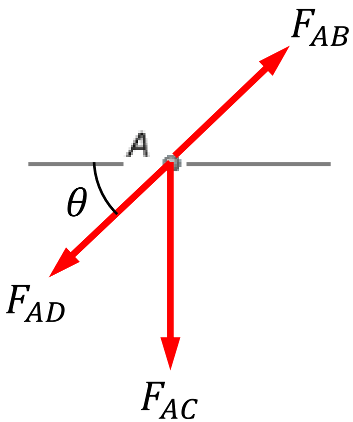
This also implies that $F_{AD} = P$, meaning member *AD* is bearing the full applied load *P*.

➢ In this case, the member *not aligned with the force* is a zero-force member.

Method of Joints – Zero-force members…

Case 3: Consider a joint that is supporting *three members*, and *two of which are aligned* to each others. And there is *no external force* applied on the joint.





Force equilibrium perpendicular to line of action of F_{AB} and F_{AD} gives

$$F_{AC}cos\theta = 0 \Rightarrow F_{AC} = 0$$

With that, it can also be inferred that

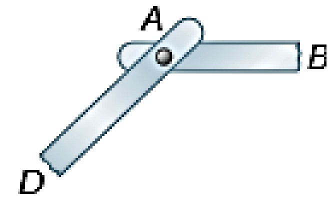
$$F_{AB} = F_{AD}$$

➢ In this case, the member that is *not aligned* is a zero-force member.

Method of Joints – Zero-force members…

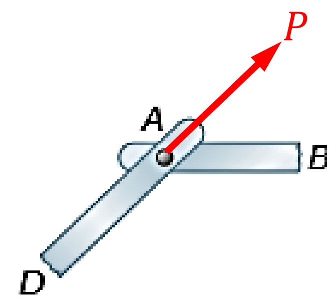
Case 1: For a joint with only *two members* with *no external loading*, and they are *not aligned* to each other.

Both the members are zero-force members.



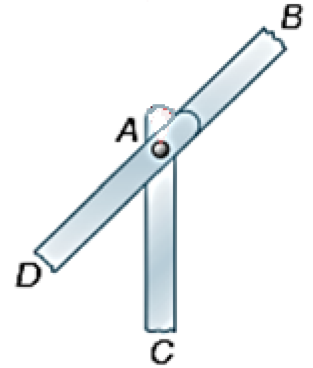
Case 2: For a joint with only *two members* and they are *not aligned* to each other. Suppose an external load is applied on the joint and its line of action is aligned with one of the member.

The one that is *not aligned to the load* is zero-force member.



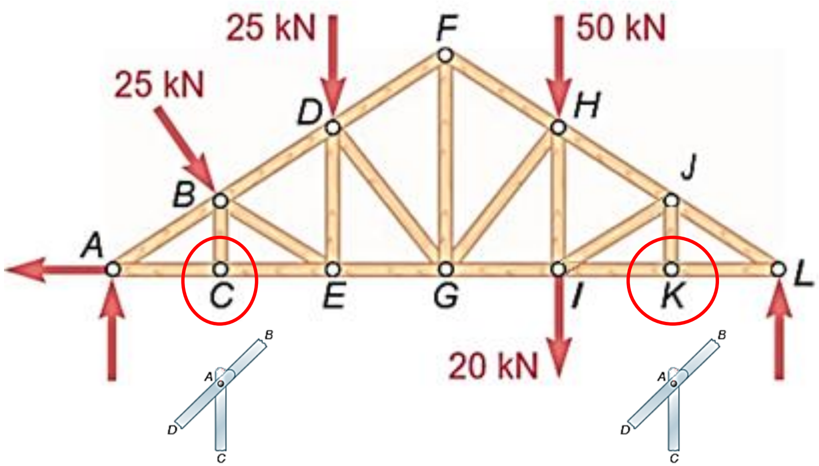
Case 3: For a joint that is supporting *three members*, and *two of which are aligned* to each others. And there is *no external force* applied on the joint.

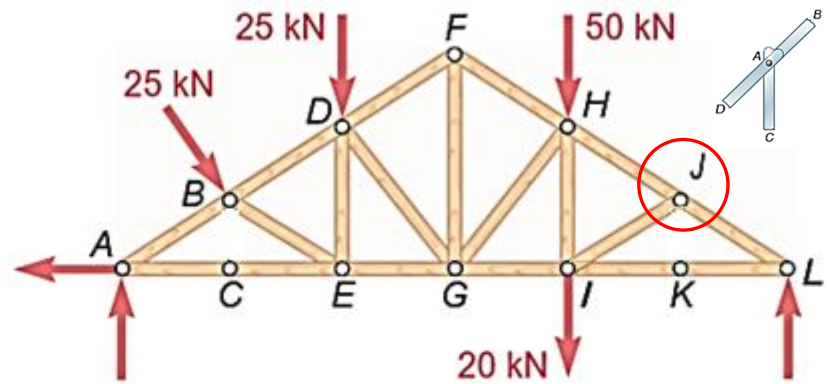
The one that is *not aligned* is zero-force member.

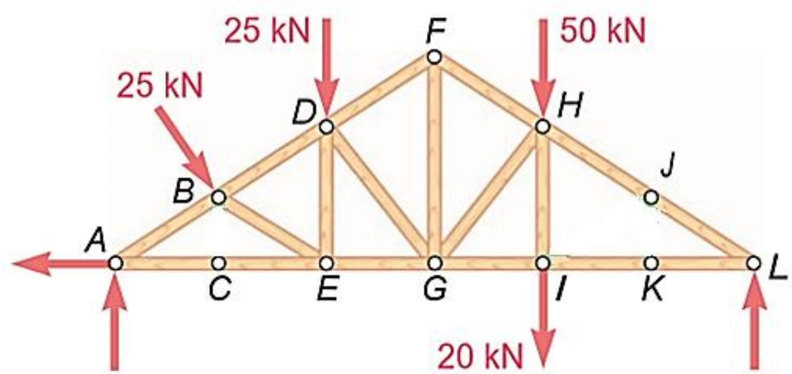


Method of Joints – Zero-force members…

- By recognizing these zero-force members under special loading conditions, we can simplify a truss analysis.



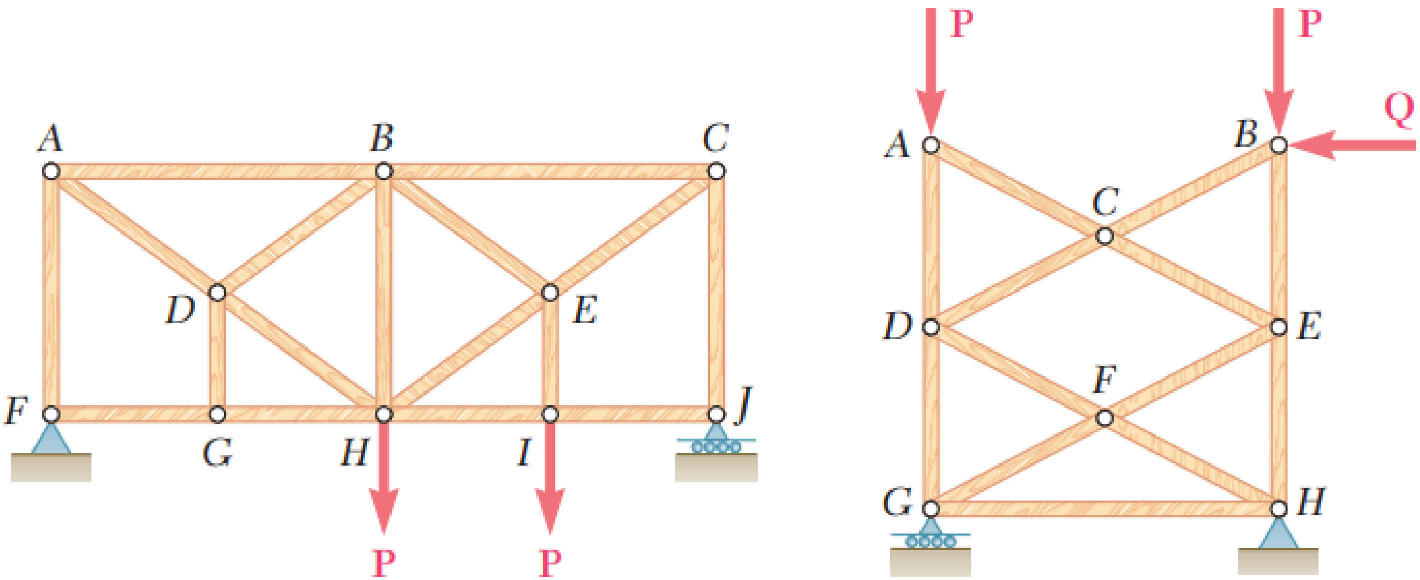




- Note that the zero-force members do not means that they are useless. It is under specific loading conditions that they become zero-force. For other loading conditions, they may have to bear loads.

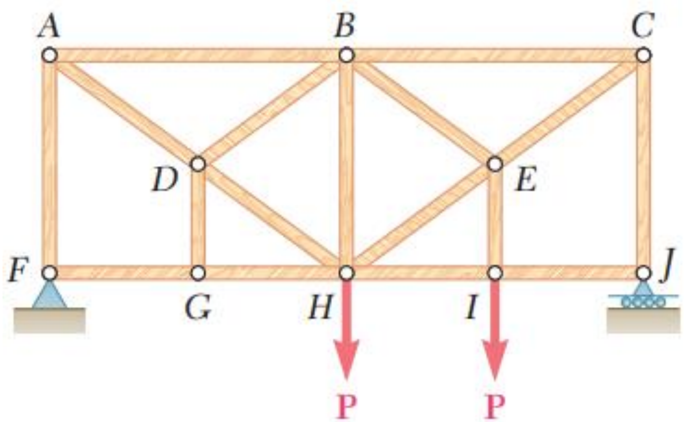
Example 3.2

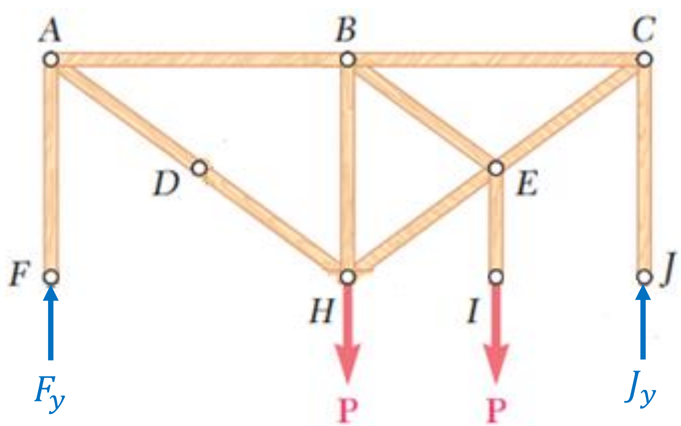
For the given loading, determine the zero-force members in each of the two trusses shown.



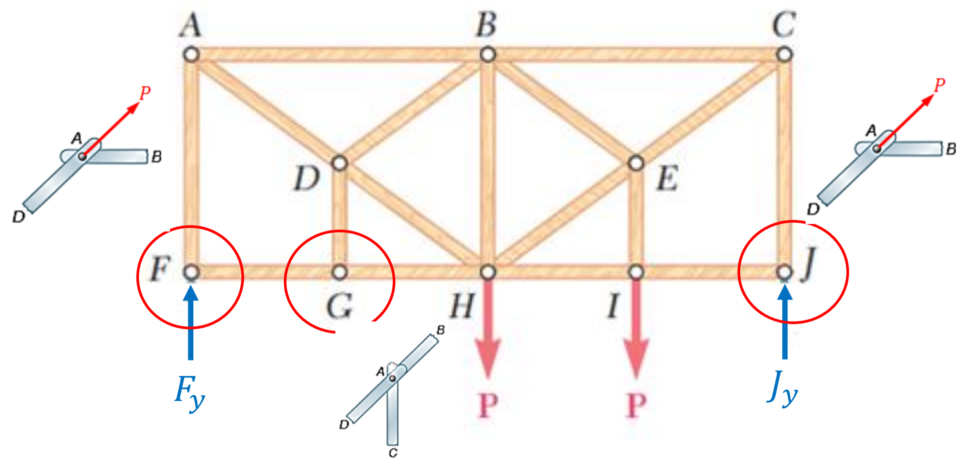
(a) (b)

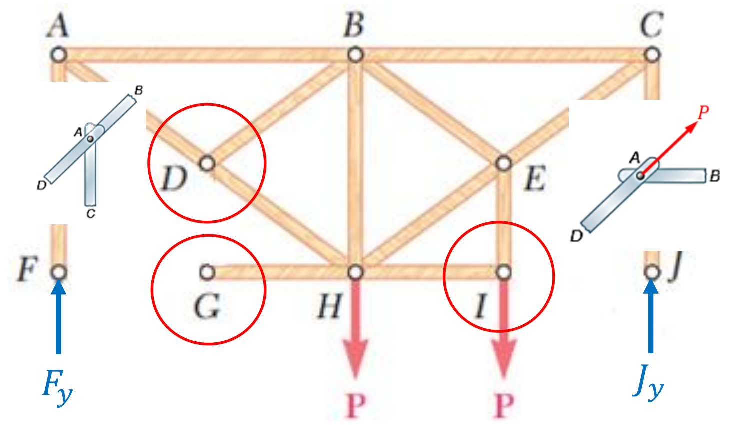
Example 3.2…





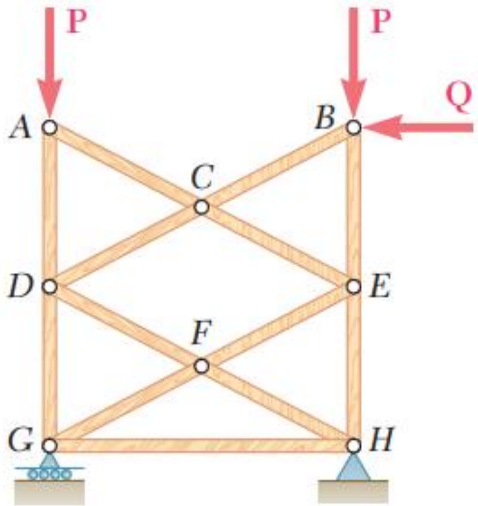
Review the truss again reveals that *GH*, *HI* and *BD* are also zero-force members.

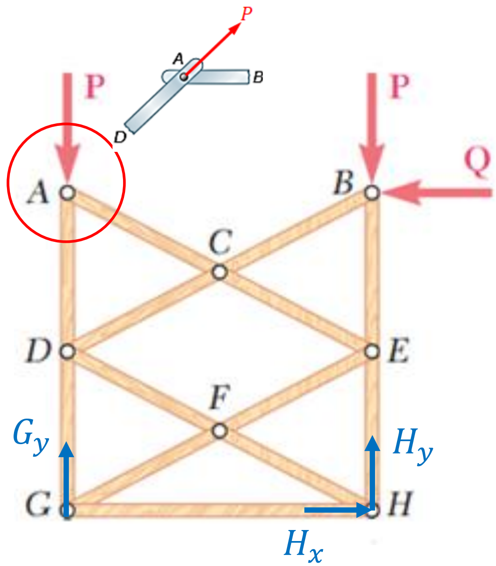


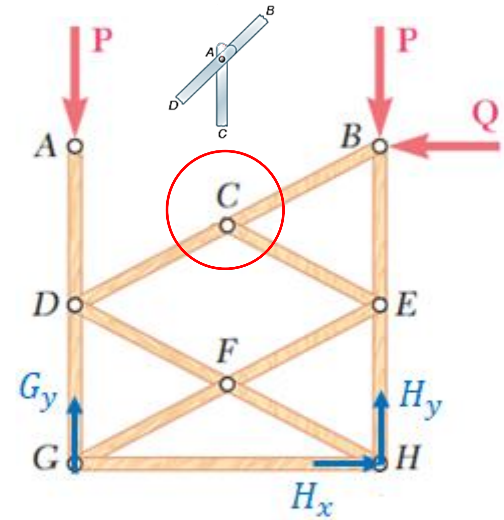


Based on the first observation, we can identify *FG*, *DG* and *IJ* are zero-force members.

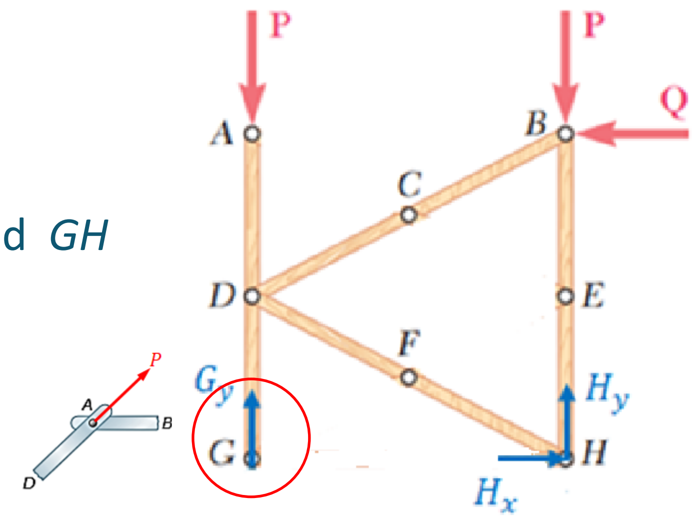
Example 3.2…

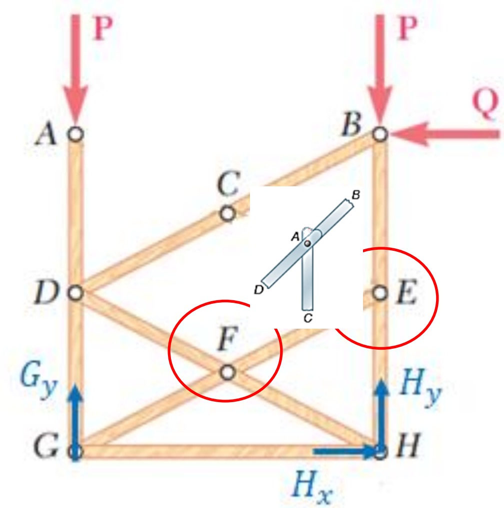






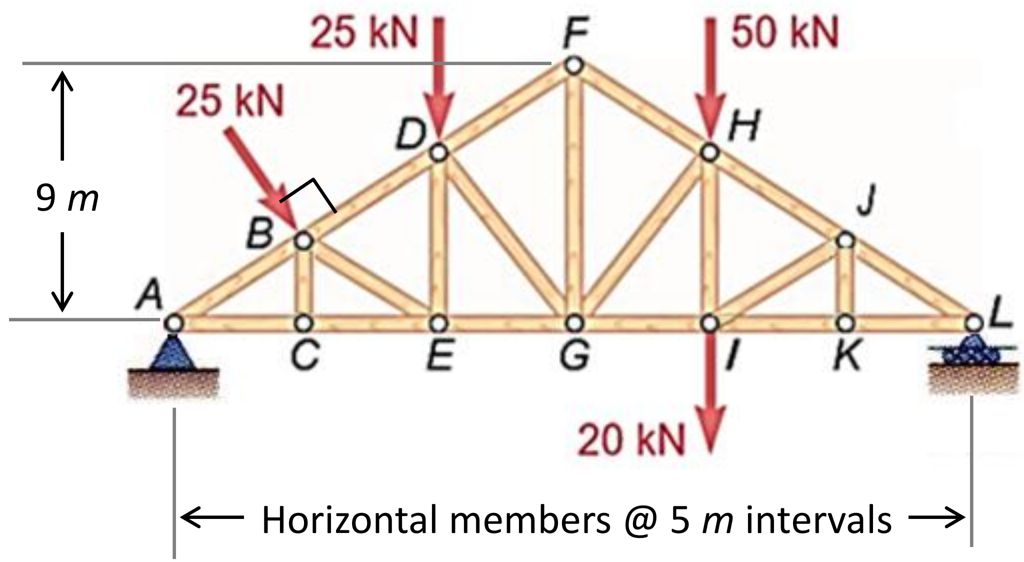
Hence, *AC*, *CE*, *EF*, *FG* and *GH* are zero-force members.



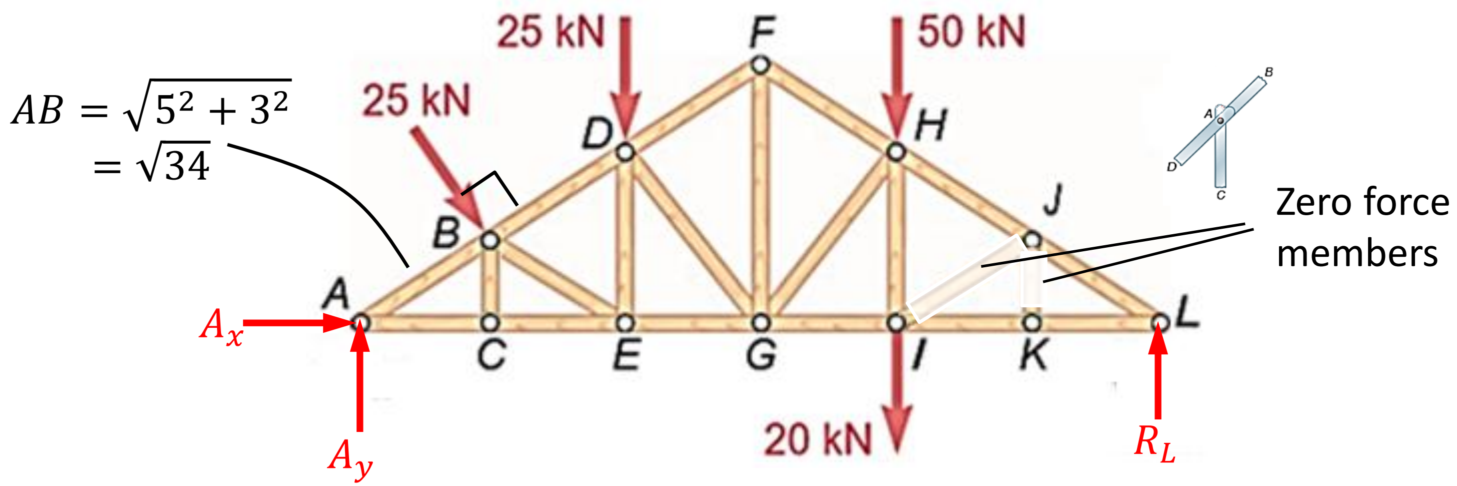


Example 3.3

Determine the forces in members *FG*, *GH*, and *HI* in the truss when it is loaded as shown.



Example 3.3…

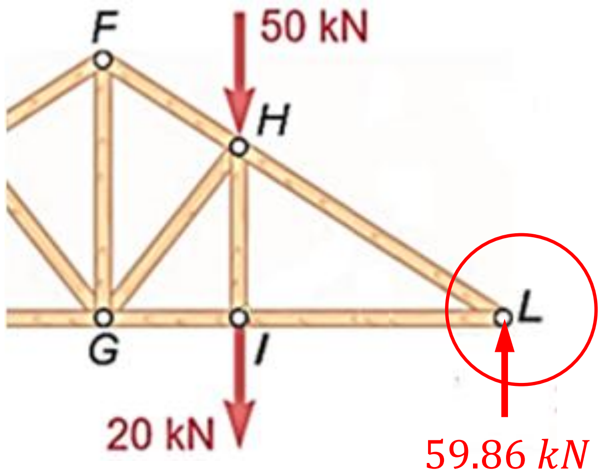


Consider the moment about *A* of entire truss, we get

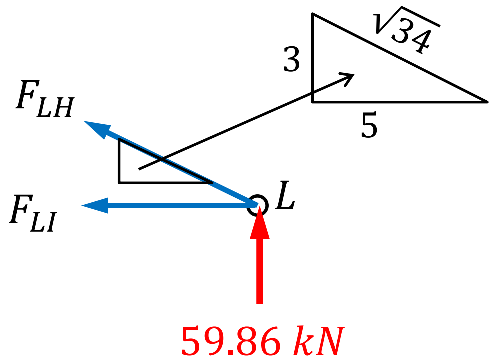
$$-(25)\left(\sqrt{34}\right) - (25)(10) - (20 + 50)(20) + 30R_L = 0$$

$$\Rightarrow R_L = 59.86\ kN$$

Example 3.3…



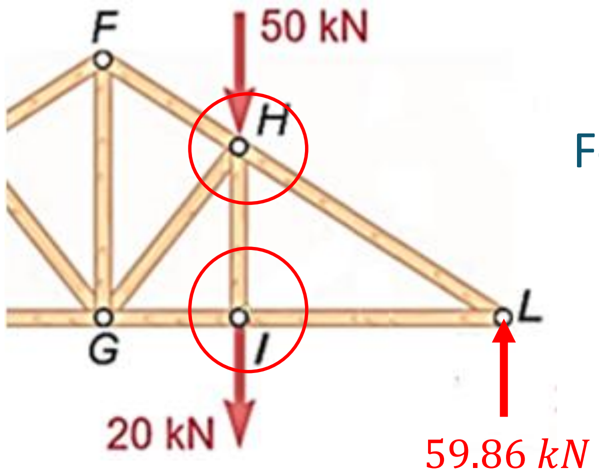
Starting with joint L,



$$\sum F_y = 0, \qquad \left(\frac{3}{\sqrt{34}}\right)F_{LH} + 59.86 = 0 \ \Rightarrow F_{LH} = -116.35\ kN$$

$$\sum F_x = 0, \qquad -F_{LI} - \left(\frac{5}{\sqrt{34}}\right)F_{LH} = 0 \ \Rightarrow F_{LI} = 99.77\ kN$$

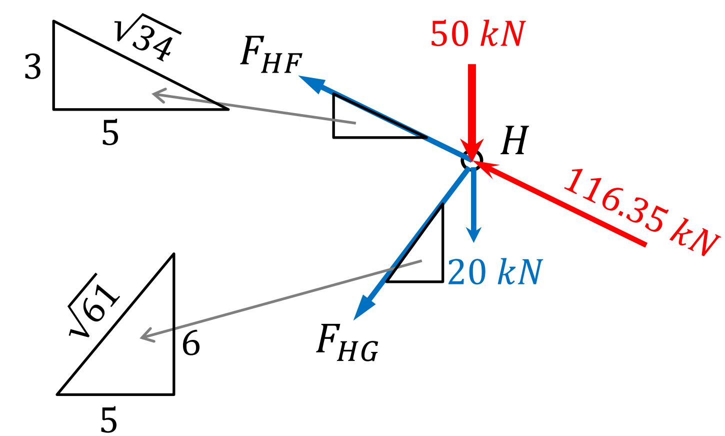
Example 3.3...



Next, moving onto joint *I*, we have

$$F_{IH} = \mathbf{20\ kN}\ (tensile)$$

Follow by joint *H*,



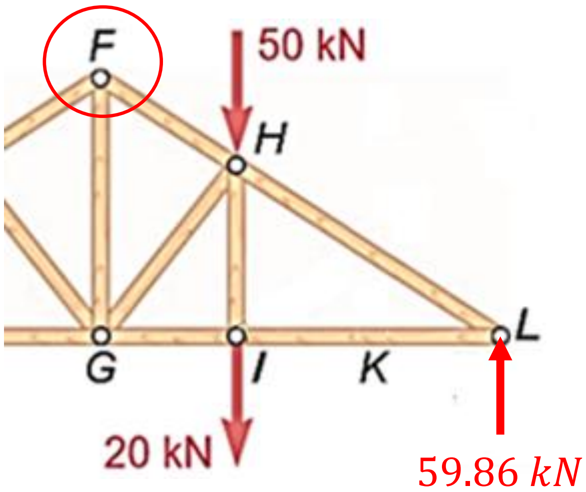
$$-\left(\frac{5}{\sqrt{34}}\right)F_{HF}-\left(\frac{5}{\sqrt{61}}\right)F_{HG}-\left(\frac{5}{\sqrt{34}}\right)(116.35)=0 \Rightarrow F_{HF}+0.7466F_{HG}=-116.35$$

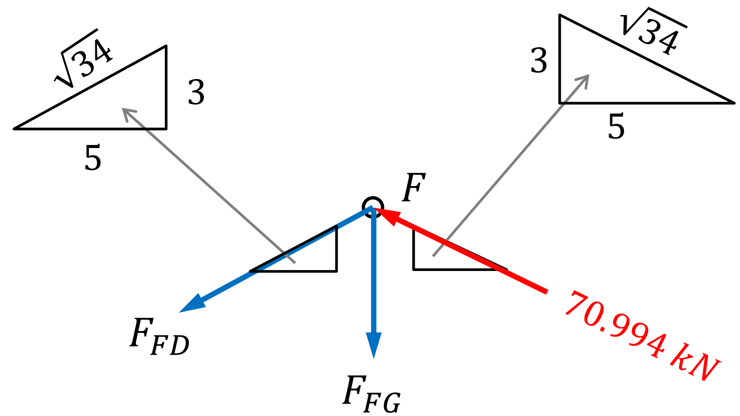
$$\left(\frac{3}{\sqrt{34}}\right)F_{HF}-\left(\frac{6}{\sqrt{61}}\right)F_{HG}-50-20+\left(\frac{3}{\sqrt{34}}\right)(116.35)=0 \Rightarrow F_{HF}-1.493F_{HG}=19.706$$

Solving the two equations, gives $F_{HF} = -70.994\ kN$ and $F_{HG} = \mathbf{-60.75\ kN}$.

Example 3.3...

Finally, consider joint *F*,





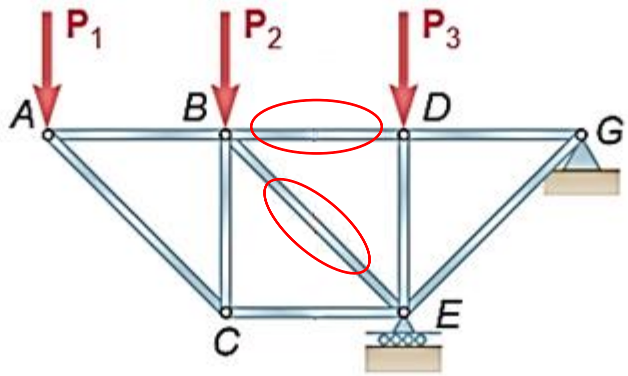
$$-\left(\frac{5}{\sqrt{34}}\right)F_{FD}-\left(\frac{5}{\sqrt{34}}\right)(70.994)=0\Rightarrow F_{FD}=-70.994\ kN$$

$$-\left(\frac{3}{\sqrt{34}}\right)F_{FD}-F_{FG}+\left(\frac{3}{\sqrt{34}}\right)(70.994)=0\Rightarrow F_{FG}=\mathbf{73.05\ kN}$$

Method of Sections

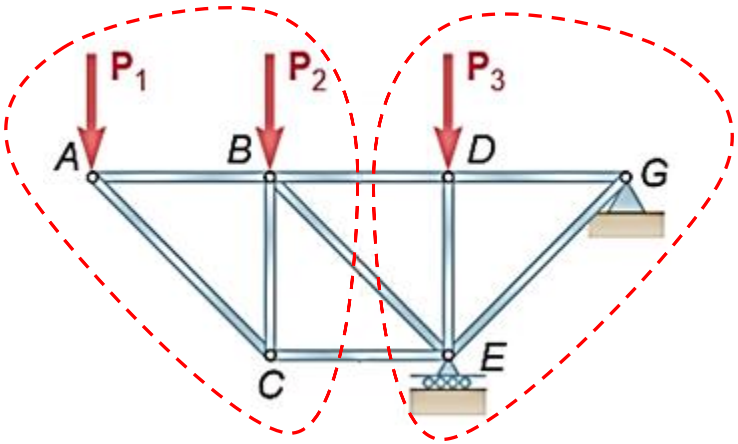
- Sometimes, only a few of the forces in the members are desired. In this case, an alternate approach called the *method of sections* will be more effective.

- Consider the truss as shown and suppose only the forces in member *BD* and *BE* is required.

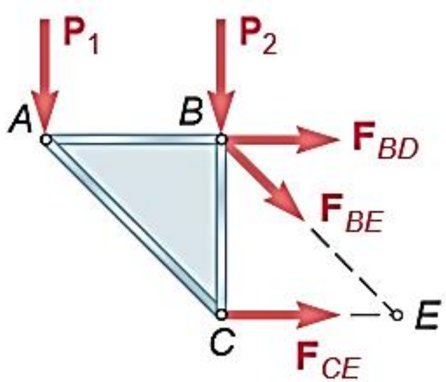


- Using the *method of joints*, you have to perform several force analyses:
  - Entire truss to solve for the reaction forces at *G* and *E*.
  - Follow by force analyses for joints *G*, *D* and *E*.
  - Or force analyses for joints *A*, *C* and *B*.

Method of Sections…



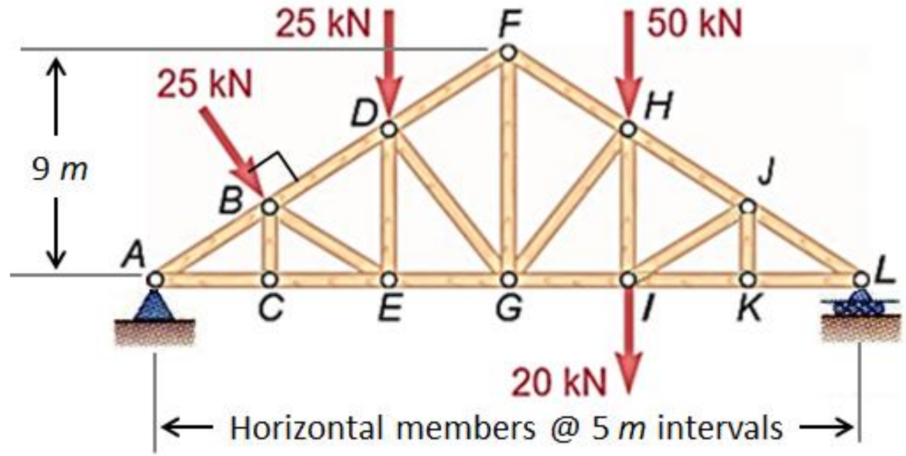
- Now, in the method of section, you need to identify part of the truss (called section) that cuts members *BD and BE* as shown.
- Note that there may be different ways to define this section. In general, there should be only be *three unknown forces* involved.



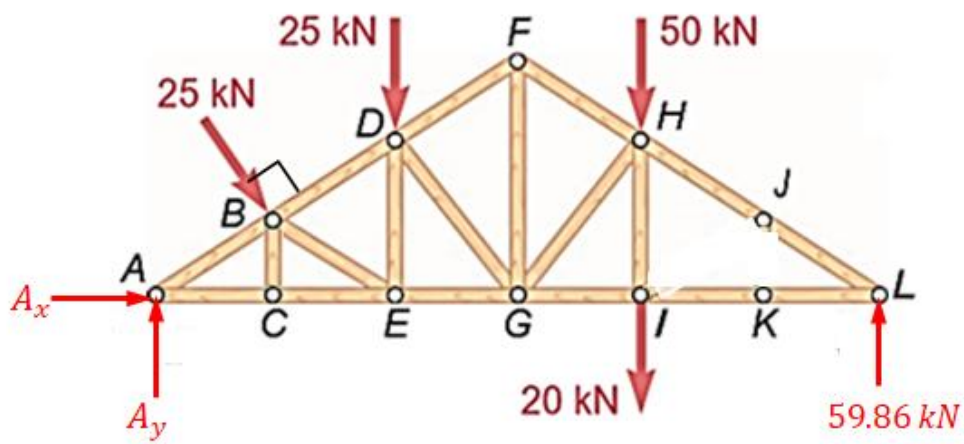
- Next, isolate the section (treated as rigid body) and impose the member forces on the joints.
- Performing the *static analysis on the section*, i.e.
  - Taking moment about *E*, we can solve for F_{BE}.
  - And by using force equilibrium in the *y* direction, we can get F_{BE}.

Example 3.4 (Redo Example 3.3)

Determine the forces in members *FG*, *GH*, and *HI* in the truss when it is loaded as shown.

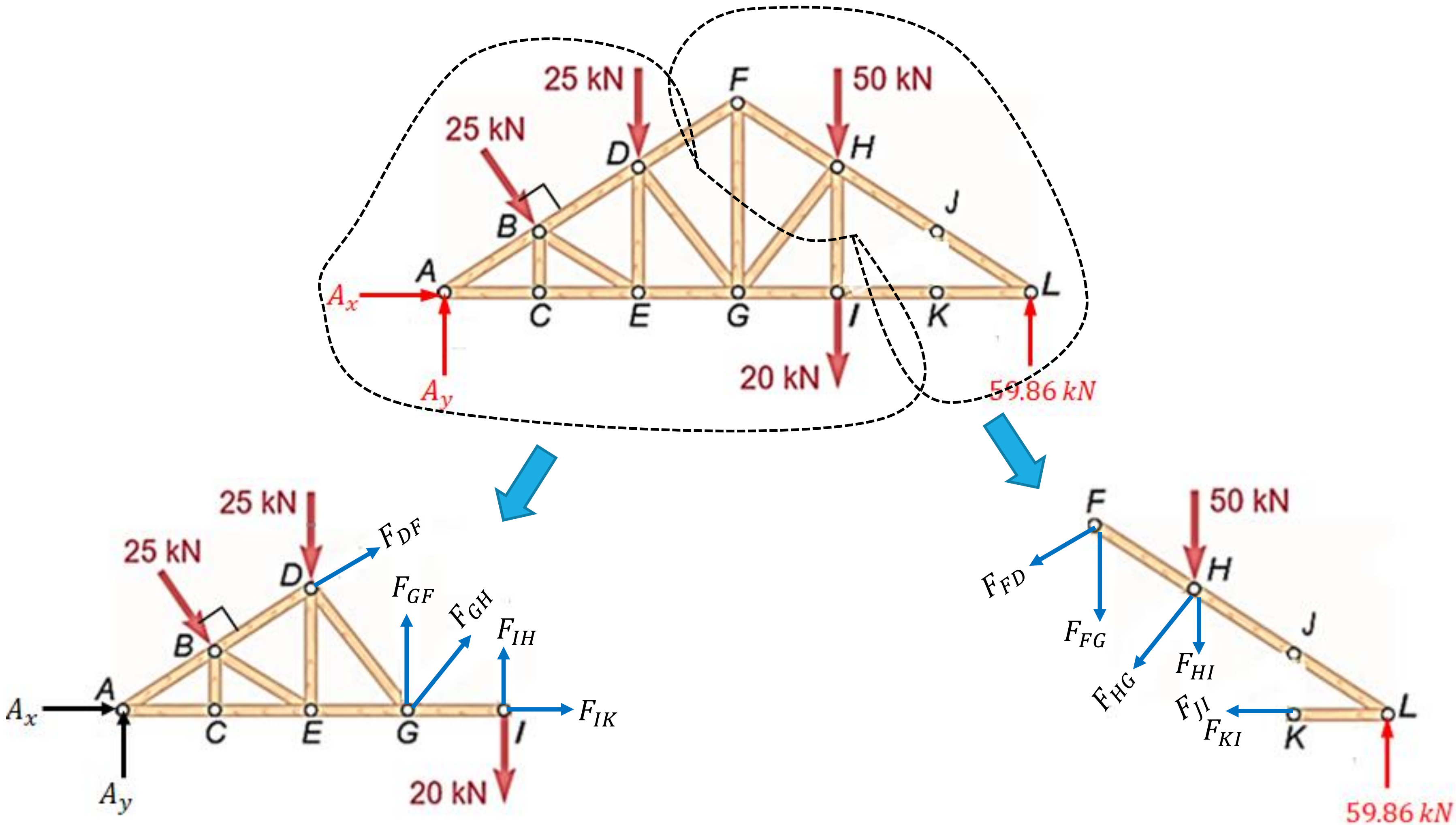


From static analysis on the entire truss, we have



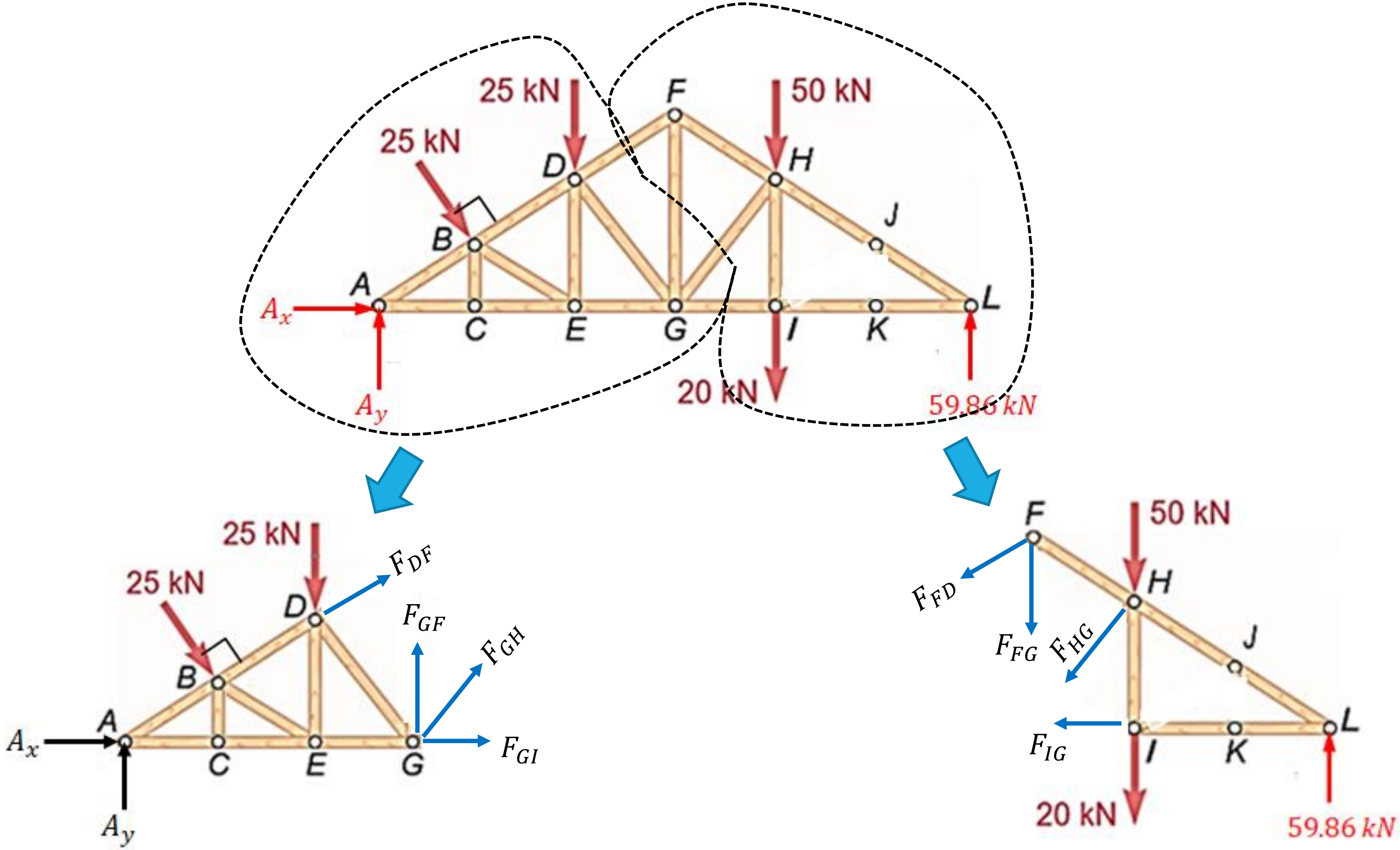
Example 3.4…

Section that cuts through all three members.



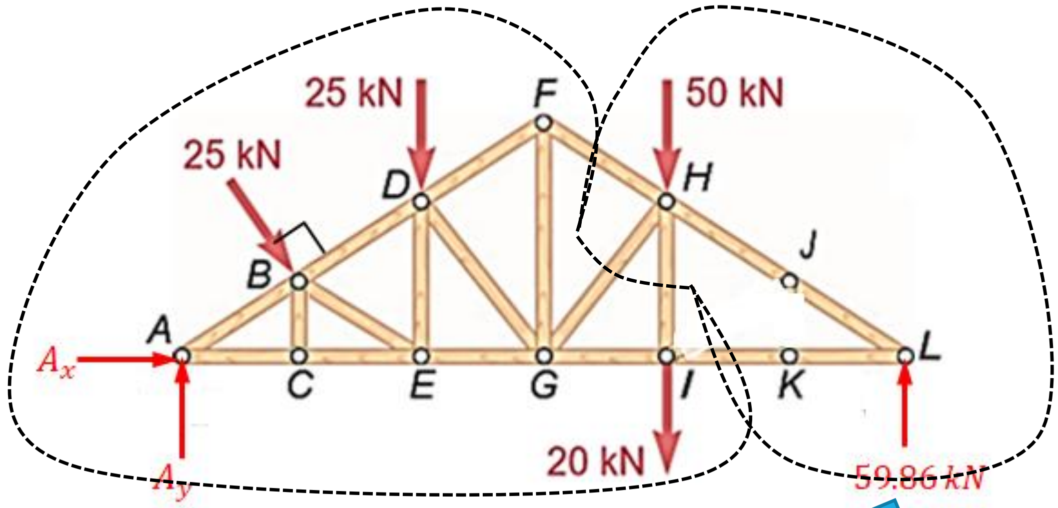
Example 3.4…

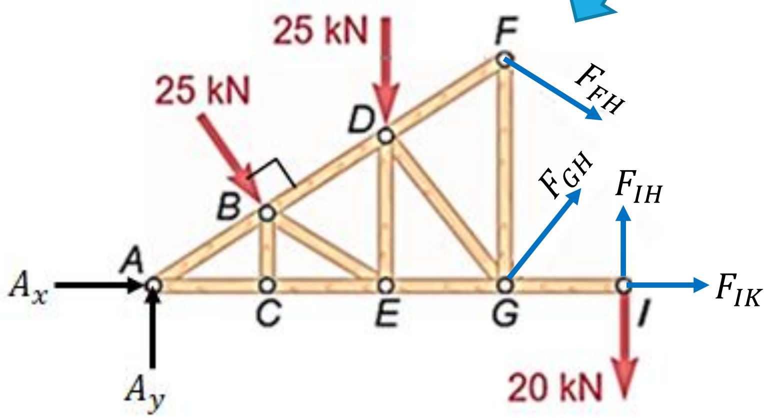
Section that cuts through two of the members.

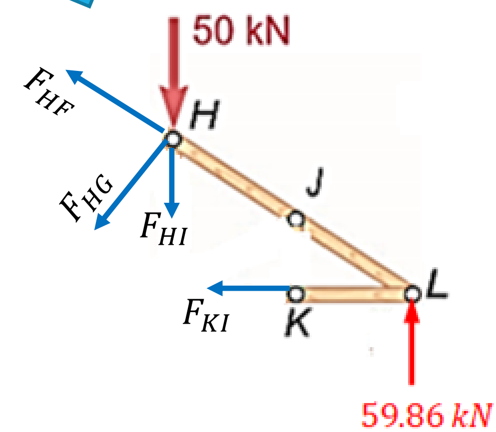


Example 3.4…

Section that cuts through another pair of members.

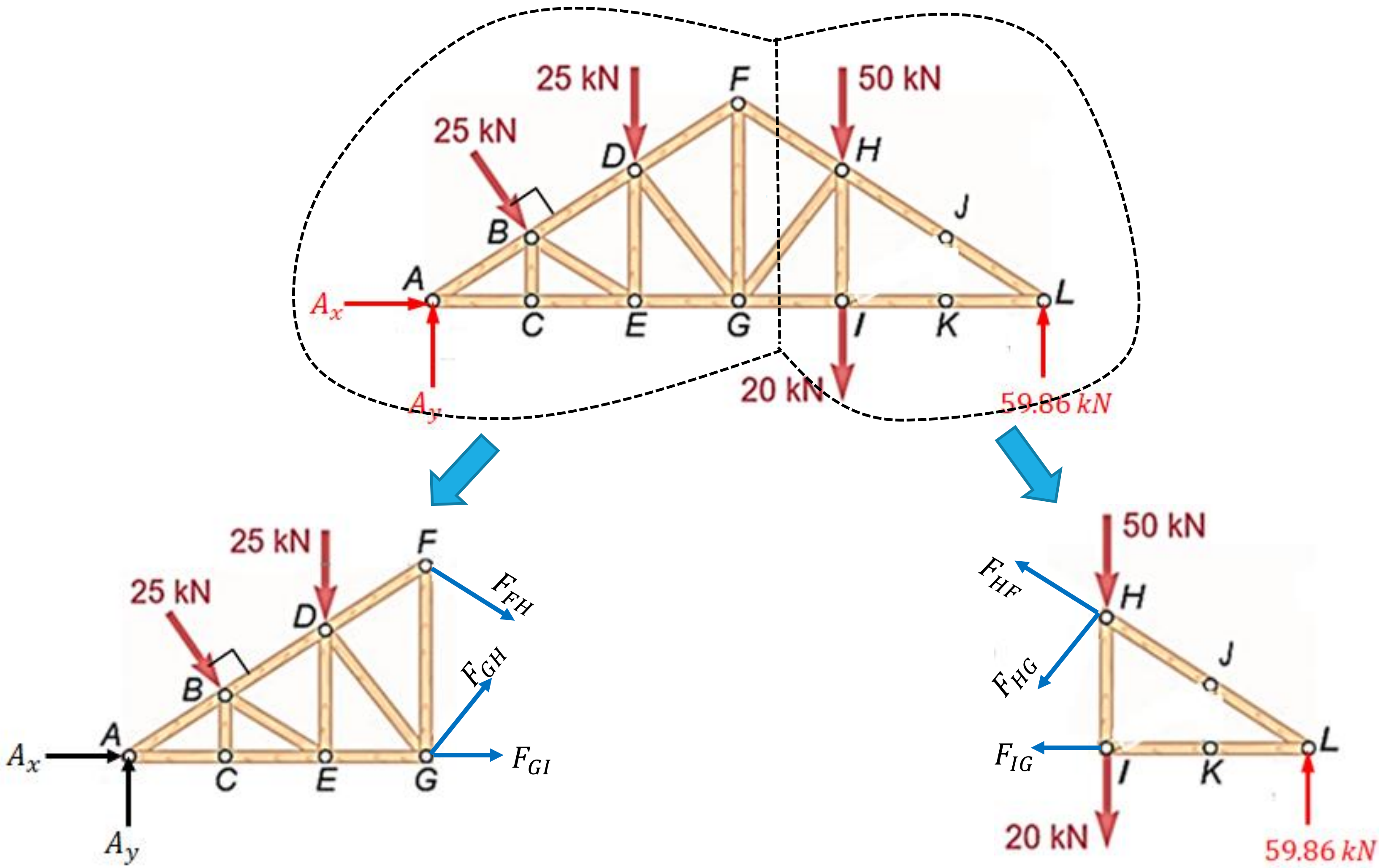




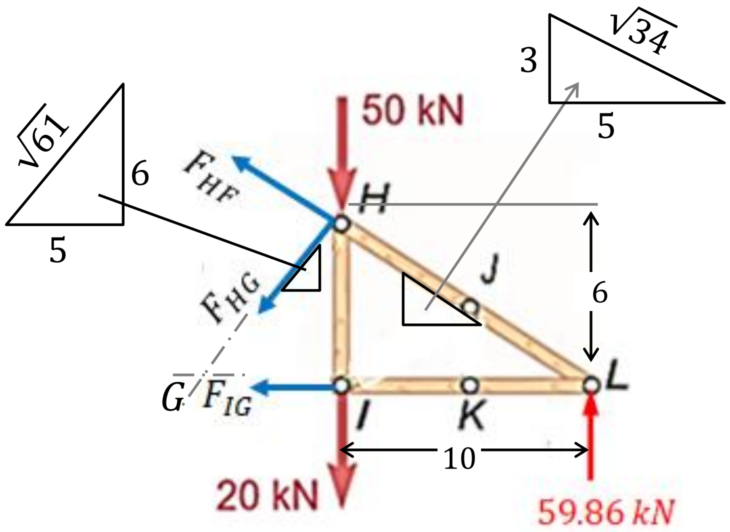


Example 3.4…

Section that cuts through one member only.



Example 3.4...



Now, consider the section defined by ΔHIL and taking moment about L, gives

$$(50+20)10+\overrightarrow{LH}\times\vec{F}_{HG}=0$$

$$\Rightarrow 700+\left((-10)\left(-\frac{6}{\sqrt{61}}F_{HG}\right)10-(6)\left(\frac{5}{\sqrt{61}}F_{HG}\right)\right)=0$$

$$\Rightarrow F_{HG}=\boldsymbol{-60.75\ kN}$$

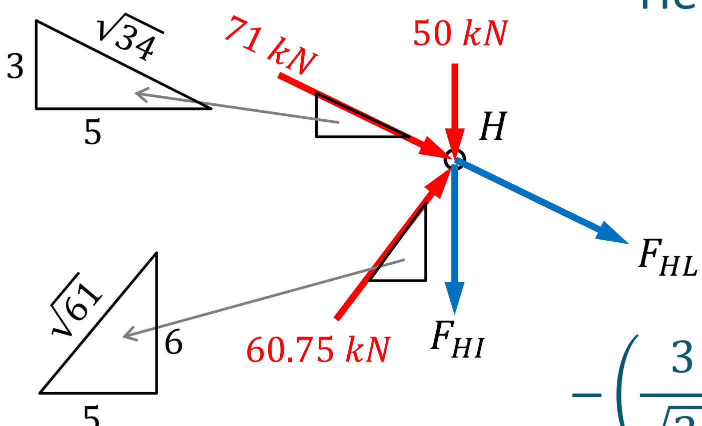
Next, consider force equilibrium in the vertical direction, gives

$$\left(\frac{3}{\sqrt{34}}F_{HF}\right)+\left(\frac{6}{\sqrt{61}}(60.75)\right)-20-50+59.86=0$$

$$\Rightarrow F_{HF}=-71.00\ kN$$

Example 3.4…

Now there only two unknowns forces at joint *H*.



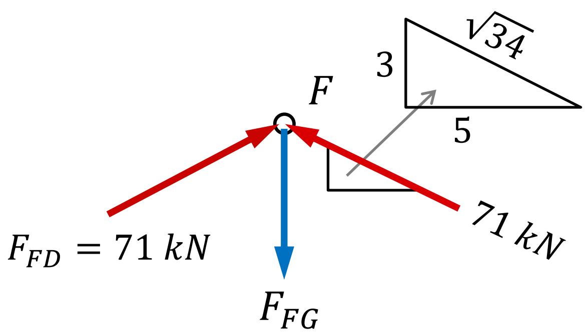
Hence, we can apply the method of joint to *H*, which gives

$$\left(\frac{5}{\sqrt{34}}\right)71+\left(\frac{5}{\sqrt{61}}\right)60.75+\left(\frac{5}{\sqrt{34}}\right)F_{HL}=0$$

$$\Rightarrow F_{HL}=-116.35\ kN$$

$$-\left(\frac{3}{\sqrt{34}}\right)71+\left(\frac{6}{\sqrt{61}}\right)60.75-50-\left(\frac{3}{\sqrt{34}}\right)(-116.35)-F_{HI}=0$$

$$\Rightarrow \boldsymbol{F_{HI}=20.0\ kN}$$



Finally, at joint *F*, its vertical force equation gives

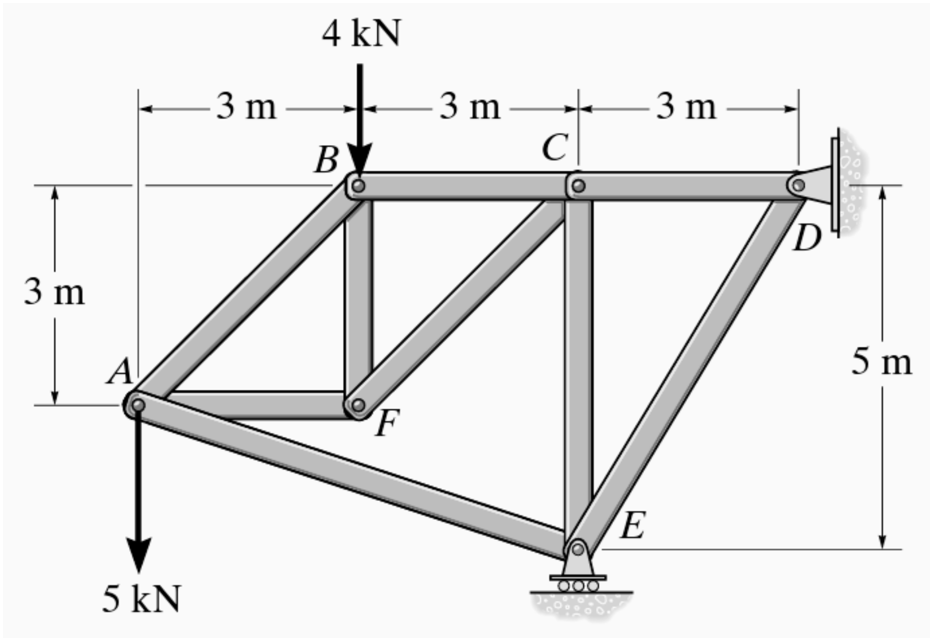
$$-F_{FG}+2\left(\frac{3}{\sqrt{34}}\right)71=0\Rightarrow F_{FG}=\boldsymbol{73.06\ kN}$$

Example 3.5

A truss that is supported at *D* and *E*, and it is loaded at joint *A* and *B* by 5 *kN* and 4 *kN*, respectively, as depicted in the figure.

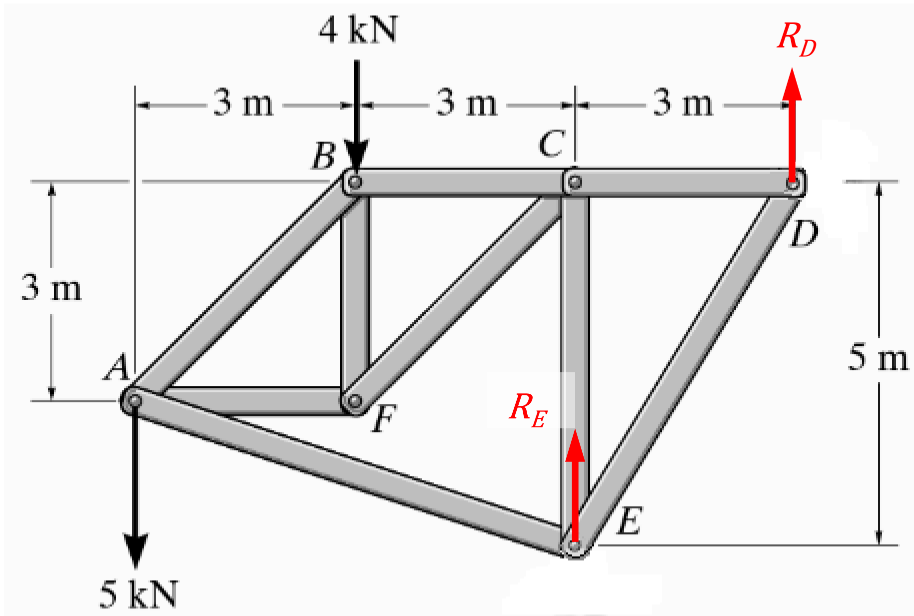
(i) Using the method of joints, determine the forces in member *CD, CE* and *DE*.

(ii) Using the method of sections, determine the force in member *AB and AF*.



Example 3.5…

Consider the entire truss. Its FBD is given as

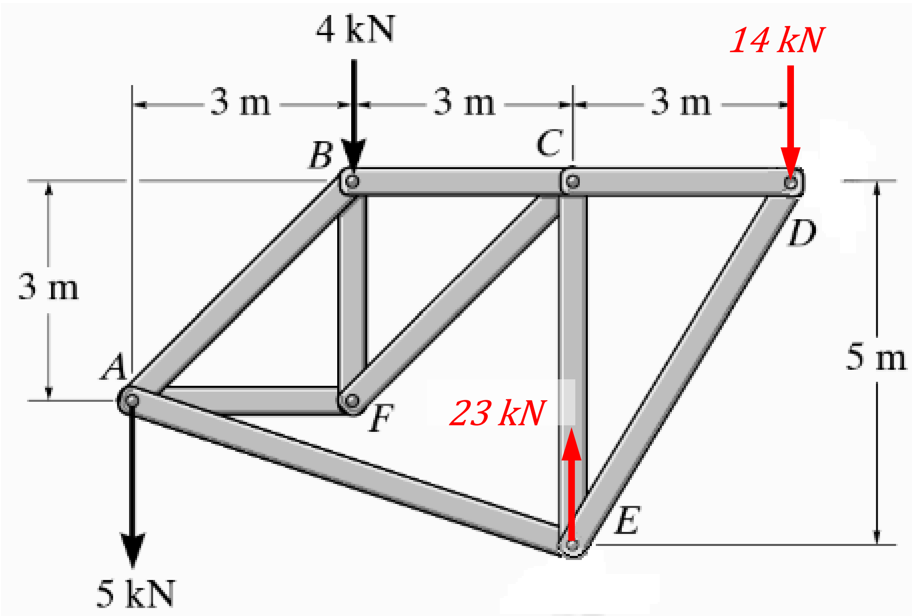


Static equilibrium of the entire truss,

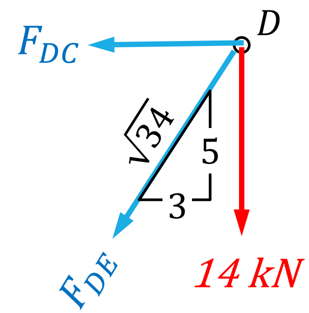
$$\sum M_D = 0\,, \qquad 4(6) + 5(9) - R_E(3) = 0 \Rightarrow R_E = 23\ kN$$

$$\sum F_y = 0, \qquad R_D + R_E - 5 - 4 = 0 \Rightarrow R_D = -14\ kN$$

Example 3.5...



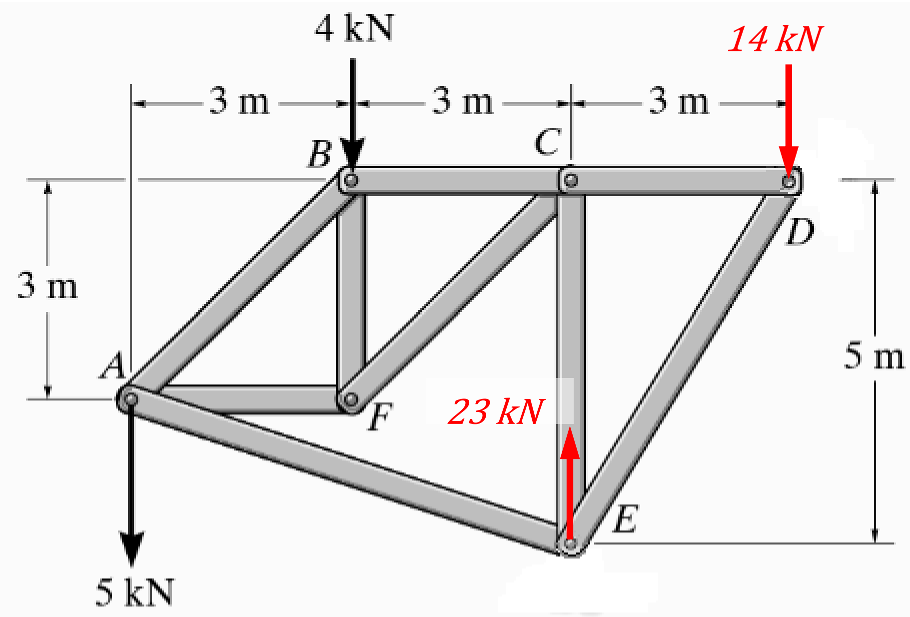
(i) Consider joint *D*,



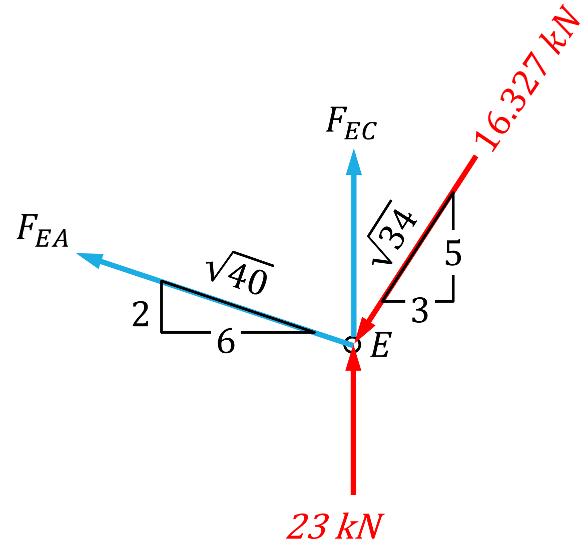
$$\sum F_y = 0, \qquad -\frac{5}{\sqrt{34}}F_{DE} - 14 = 0 \;\Rightarrow\; F_{DE} = \mathbf{-16.327\ kN}$$

$$\sum F_x = 0, \qquad -F_{DC} - \frac{3}{\sqrt{34}}F_{DE} = 0 \;\Rightarrow\; F_{DC} = -\frac{3}{\sqrt{34}}(-16.327) = \mathbf{8.400\ kN}$$

Example 3.5...



To solve for force in *CE*, we use joint *E* since it has only two unknowns.

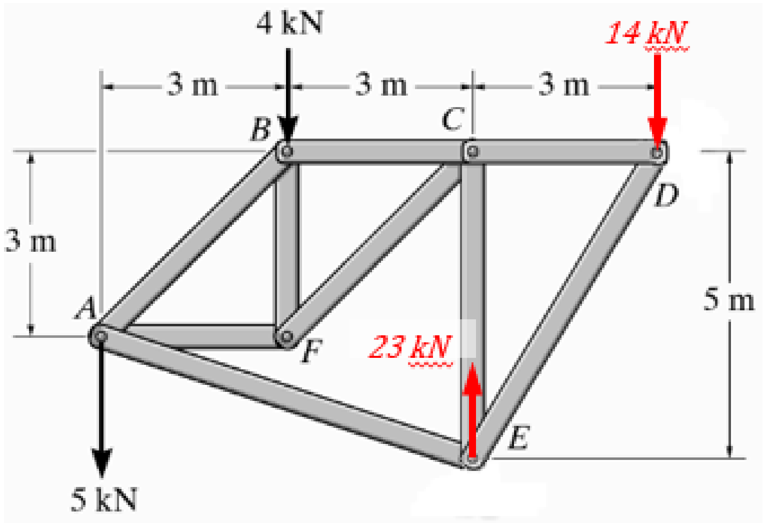


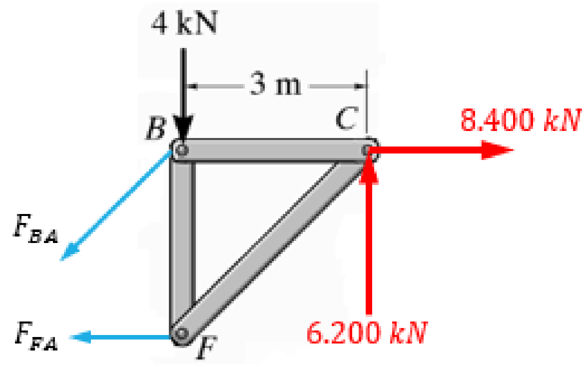
$$\sum F_x = 0, \qquad -\frac{6}{\sqrt{40}}F_{EA} - \frac{3}{\sqrt{34}}(16.327) = 0 \;\Rightarrow\; F_{EA} = -8.8544\ kN$$

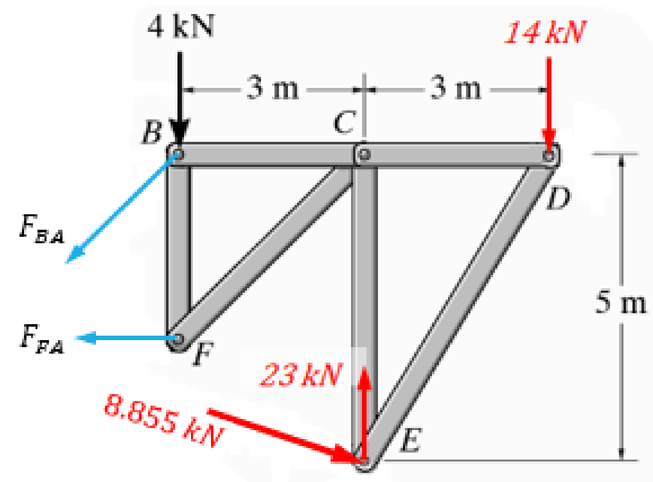
$$\sum F_y = 0, \qquad F_{EC} + \frac{2}{\sqrt{40}}F_{EA} - \frac{5}{\sqrt{34}}(16.327) + 23 = 0 \Rightarrow F_{EC} = \mathbf{-6.200\ kN}$$

Example 3.5…

(ii) To determine the forces in member *AB* and *AF* using method of section, we can consider the following sub-sections.

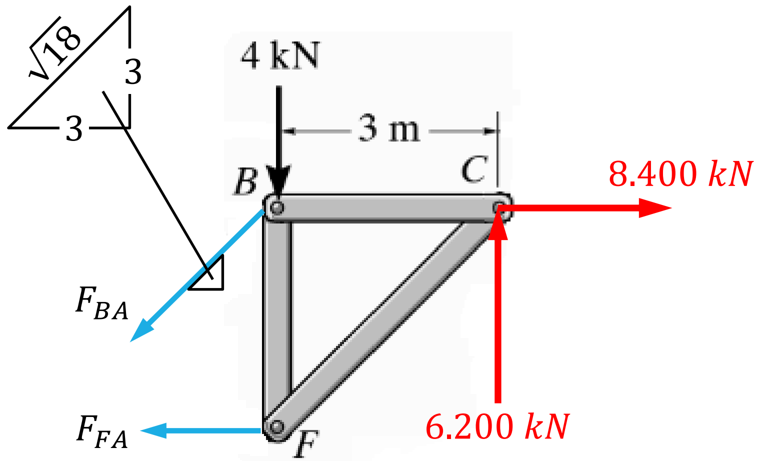






Example 3.5...

For simplicity, we should *BCF.*



Using the force equilibrium, gives

$$\sum F_y = 0\,, \qquad -\frac{3}{\sqrt{18}}F_{BA} - 4 + 6.200 = 0$$

$$\Rightarrow F_{BA} = \mathbf{3.111\ kN}$$

$$\sum F_x = 0\,, \qquad -F_{FA} - \frac{3}{\sqrt{18}}(3.111) + 8.400 = 0$$

$$\Rightarrow F_{FA} = \mathbf{6.200\ kN}$$

Alternatively, taking moment about *B*, gives

$$-F_{FA}(3) + 6.200(3) = 0 \Rightarrow F_{FA} = \mathbf{6.200\ kN}$$

And force equilibrium in the vertical direction, gives

$$-\frac{3}{\sqrt{18}}F_{BA} - 4 + 6.200 = 0 \Rightarrow F_{BA} = \mathbf{3.111\ kN}$$

Example 3.6

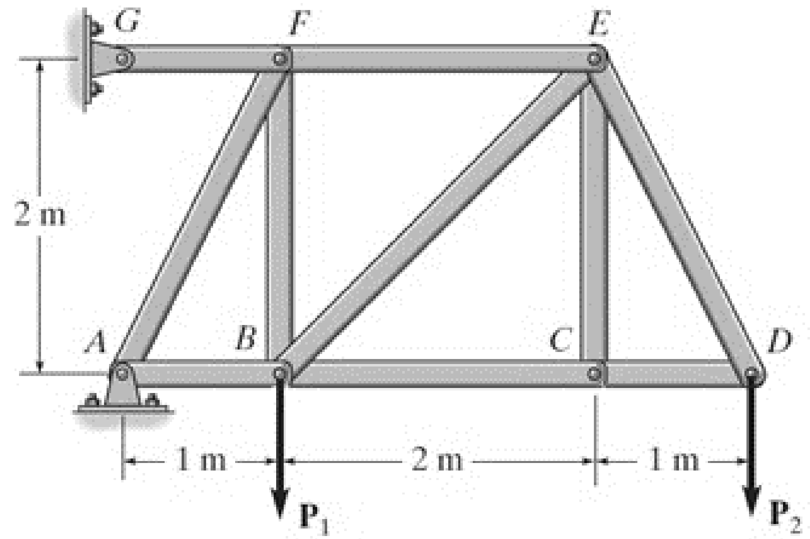
For the truss shown in the figure, determine the followings:

(i) Given that the maximum compressive load in member *CD* is 6 *kN*, while the maximum tensile load in member *DE* is 15 *kN*, what is the corresponding maximum load P_2?

(ii) Now, given that P_1 = 10 *kN* and P_2 = 8 *kN*, determine the reaction forces at the supports *A* and *G*. (*8 marks*)

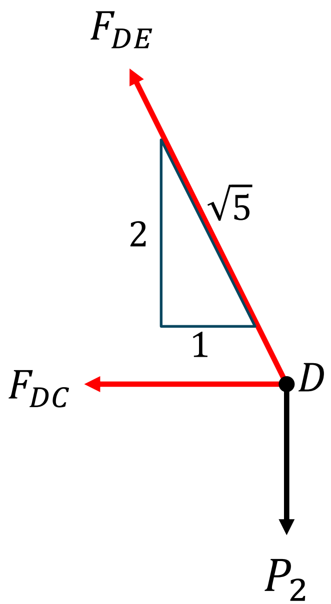
(iii) Finally, using the method of section only, determine the forces in members *EF*, *BE* and *BC*. State clearly whether they are in tension or compression.

Note: Include all the FBDs in your solutions.



Example 3.6...

(a) Members *CD* and *DE* are connected at joint *D*. Hence, by applying the method of joint at D, we have



Force equilibrium, gives

$$\sum F_x = 0, \qquad \left(\frac{1}{\sqrt{5}}\right) F_{DE} + F_{DC} = 0$$

$$\sum F_y = 0, \qquad \left(\frac{2}{\sqrt{5}}\right) F_{DE} - P_2 = 0$$

When F_{DE} reaches its maximum limit of 15 *kN*,

$$F_{DC} = -\left(\frac{1}{\sqrt{5}}\right)(15) = -6.71\ kN \text{ and } P_2 = \left(\frac{2}{\sqrt{5}}\right)(15) = 13.42\ kN$$

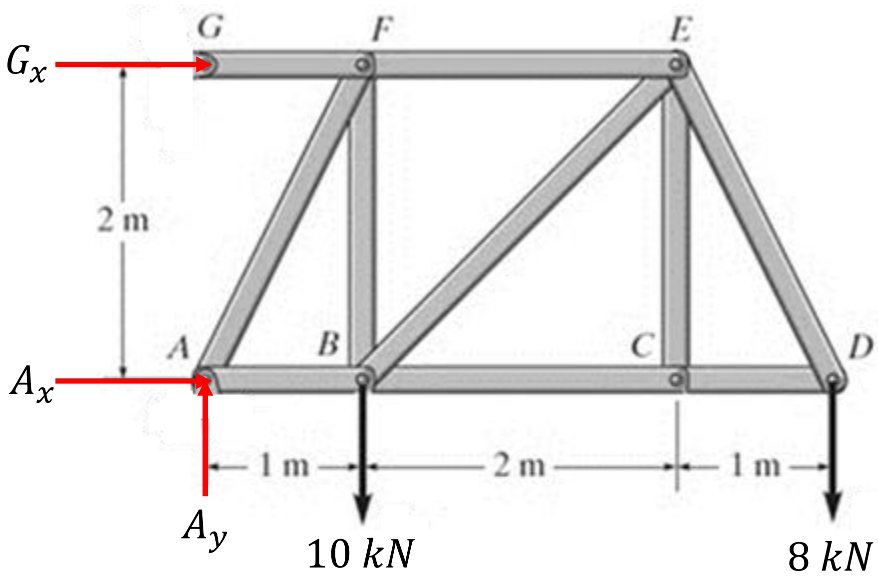
When F_{DC} reaches its maximum limit of -6 *kN*,

$$F_{DE} = -\sqrt{5}(-6) = 13.42\ kN \text{ and } P_2 = \left(\frac{2}{\sqrt{5}}\right)(13.42) = 12.0\ kN$$

Comparing the two results, $F_{DC} = -6\ kN$ is the limiting case. Hence, $P_{2,max} = \mathbf{12\ kN}$.

Example 3.6…

(b) Consider the entire truss.



Although joint G is constrained by a pin, which implies that it can potentially induce two reaction forces, member GF is a 2-force body.

As such, the reaction force is expected to be inline with member GF, which is horizontal.

Taking moment about A, gives

$$-G_x(2) - 10(1) - 8(4) = 0$$
$$\Rightarrow G_x = \mathbf{-21\ kN}$$

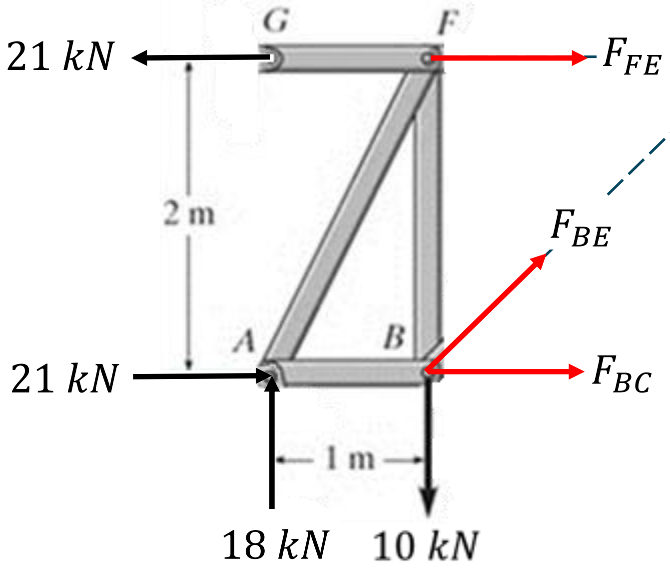
And force equilibrium, gives

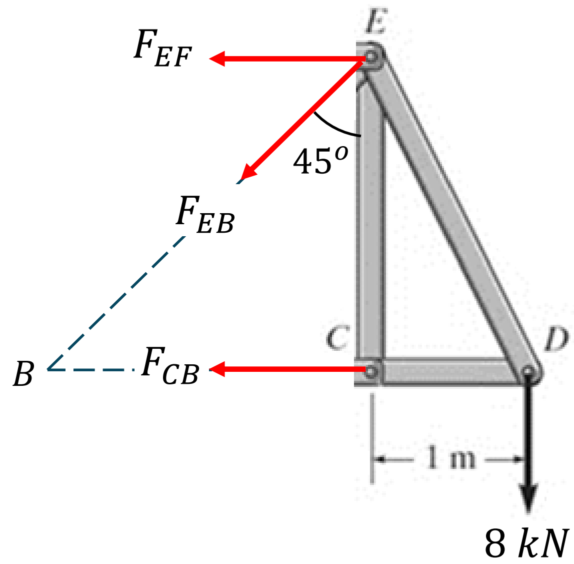
$$A_x - G_x = 0 \Rightarrow A_x = \mathbf{21\ kN}$$

$$A_y - 10 - 8 = 0 \Rightarrow A_y = \mathbf{18\ kN}$$

Example 3.6…

(c) Based on member *EF*, *BE* and *BC*, two sections can be used here.





$$\sum M_E = 0, \qquad -F_{CB}(2) - 8(1) = 0 \Rightarrow F_{CB} = \mathbf{-4\ kN}$$

$$\sum F_y = 0, \qquad -F_{EB}cos45 - 8 = 0 \Rightarrow F_{EB} = \mathbf{-11.32\ kN}$$

$$\sum F_x = 0, \qquad -F_{EF} - F_{EB}sin45 - F_{CB} = 0 \Rightarrow F_{EF} = \mathbf{12.0\ kN}$$